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Making more with less: confinement effects for more sustainable chemical transformations

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Chemical reactions require close physical contact between reactants, catalysts, and intermediates to proceed. In a bulk homogeneous solution, this is often governed by random collisions based on molecular diffusion. A powerful concept is the use of confinement to physically restrict the reacting species within the same physical space, increasing the likelihood of successful collisions. Controlling the identity and geometry of the co-located reactants also provides the opportunity to impart improved regio- and chemoselectivity for these processes. This effect has been explored in a number of contexts, including catalytic performance, yet its potential in providing greener processes has not been fully recognized. This review discusses different modes of confinement that have been explored for catalytic applications grouped as confinement within pores and cavities, confinement by encapsulation within hollow solids, and confinement within self-assembled systems. The theory behind changes in reaction outcomes due to confinement and selected examples of reactions where confinement has played an important role are discussed. We highlight the important implications of confinement effects towards the discovery of more sustainable processes as well as the relevance of these outcomes to the field of green chemistry. Key challenges and future directions within the field are highlighted.

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1. Introduction

Conventionally, chemical solutions are viewed as continuous, homogenous, and isotropic systems with a random distribution of solutes in the solution. Yet, many important chemical phenomena occur outside of this idealized view. For instance, in liquid crystals, colloidal suspensions, and supramolecular arrangements, anisotropy is central to understanding effects from the molecular to macroscopic scales.^{1–3} The ability to influence the distribution and conformation of molecules and ions at the nanoscale through their confinement within certain spaces provides the opportunity to impart greater control over chemical reactions and their outcomes. Such control has the potential to markedly improve chemical selectivity, reduce waste,

increase reaction rates, and reduce energy requirements, highlighting the great potential of this approach from a green chemistry perspective. Herein, we refer to confinement as the spatial restriction of reactants, intermediates, and catalysts within structures of nanoscale and sub-nanoscale dimensions. Such restrictions result in unique adsorption and diffusion properties, which cause the confined species to behave differently from what they would in a homogeneous, isotropic environment. Confinement effects can allow for a local increase in the concentration of reactants as well as influence their interaction with catalytic sites. Combined, these effects can be used by chemists to improve processes and move towards meeting the principles of green chemistry.⁴ As will be exemplified in this review, confined systems can enable, for instance, higher selectivities (waste prevention, principle 1), less excess of reactants (atom economy, principle 2), lower use of highly reactive species (less hazardous synthesis, principle 3), milder reaction conditions (energy efficiency, principle 6), improved catalytic properties (catalysis, principle 9) and, overall, safer chemical processes (safer chemistry, principle 12) (Scheme 1). Hence, by introducing physical spatial constraints, chemists can make more with less, and effectively deliver on the promise of greener chemical processes.

Confinement effects have played an essential role in chemical reactivity since the beginning of life itself. Nature, a renowned inspiration to green chemists, relies on confinement effects to carry out numerous chemical processes inside enzy-

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matic cavities, which would likely not happen otherwise. However, the concept of confinement was only introduced in catalysis in the 1980s with the seminal work of Eric Derouane *et coll.*^{5–8} They established that molecules inside zeolite pores tend to optimize their van der Waals interactions with the walls and experience magnified physisorption energetics because of the atomic-range curvature inside the pores. Such effects reflect in the diffusion properties of reactants and intermediates inside the pores and directly influence reaction rates. The work of Derouane on zeolites is ubiquitous when studying confinement effects, and it has already been excellently reviewed.^{9,10}

The concept of confinement has now expanded beyond zeolites, and it has also been observed inside the porous channels of mesoporous silica^{11–13} and carbon nanotubes,^{14–16} for instance. Such materials feature pores that are connected to the bulk reaction medium, generating gradients of diffusion rates that play important roles in the reactivity of substrates.^{17,18} In addition, different encapsulation strategies have also emerged to confine chemical reactions. In this review, encapsulated reactions will differ from the ones inside porous structures in the sense that they will refer to closed, discrete nanoreactors rather than pores contained in continuous solid networks. The encapsulation of reactions includes strategies with rich organic and inorganic chemical compositions and architectures ranging from hollow to multi-shelled particles.¹⁹ Apart from solid cages, supramolecular chemists have been able to create self-assembled structures which take advantage of the organization of molecules in solution to also carry out spatially confined reactions.^{20–22}

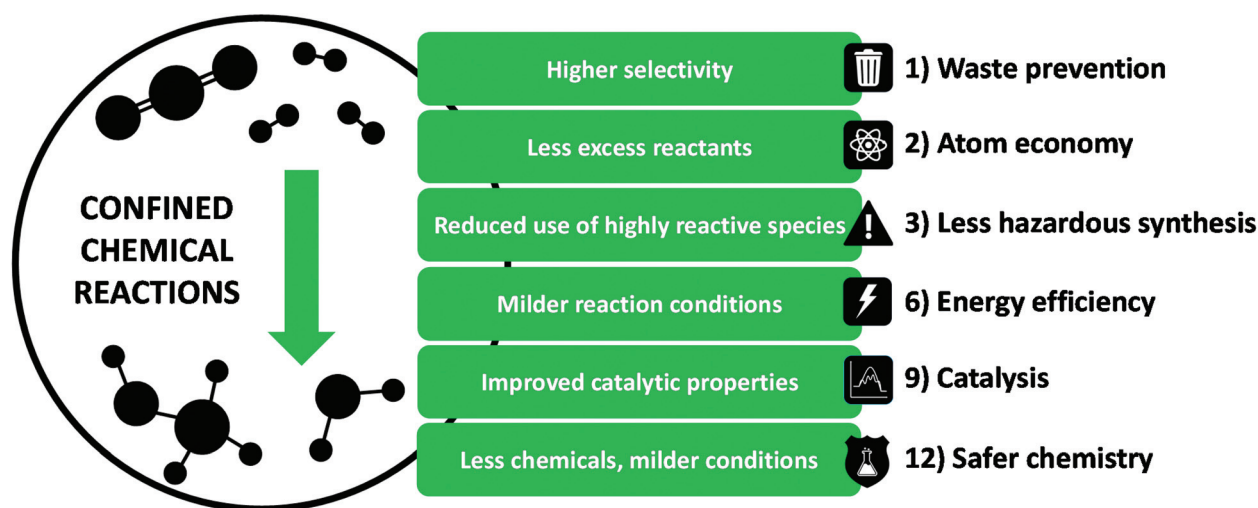
The effects of confinement have already been reviewed in different contexts, including an enlightening review by Goettmann & Sanchez in 2007²³ and more recent ones that highlighted synergic effects,²⁴ supramolecular

arrangements,^{25,26} and a broad and comprehensive book edited by Poli and published in 2017.²⁷ Here we do not intend to exhaustively review the literature on confinement effects, but we will rather explore it from a green chemistry perspective. We will be highlighting recent creative studies with the aim of introducing spatial confinement into the green chemistry toolbox as an approach for chemists to make more products from fewer resources. We also focused our attention on examples where confinement was employed as a means to deliver better processes in strategic sustainable areas such as energy, biomass, or pollution remediation.

We will begin by reviewing recent theoretical and mechanistic studies that provide evidence for the role of spatial confinement. Following, we will present selected works from the literature, which are organized in three major sections according to the nature of the confining medium in each specific case: confinement in pores, confinement by encapsulation, and confinement in self-assembled structures.

2. Theoretical studies of confinement

While it has been experimentally observed for over three decades that chemical reactivity can be influenced by the introduction of spatial constraints, it is intrinsically challenging to experimentally isolate the various factors that can account for it. By using the reactive Monte Carlo simulation method and transition state theory, the Gubbins group studied confinement in several reactions, which they have compiled in a series of review articles.^{28–30} The group pioneered computational studies on confinement effects, showing that both the equilibrium position and dynamics of chemical reactions can be affected by spatial restrictions. Equilibria can be shifted by two



Scheme 1 Various ways confinement effects can be used to apply green chemistry principles. This scheme is a simplification of the main principles related to each improvement observed throughout the examples that will be highlighted in this review, but each improvement might be related to more than one principle and *vice versa*.

main mechanisms: firstly, an increase in the density of molecules inside of the confined space can favour reactions that decrease the number of molecules of the system; secondly, equilibria can also change as a result of selective adsorption to the walls, which will change the effective concentration of the adsorbed component in the reaction media. Much of the recent literature, however, has been dedicated to the study of what the Gubbins group referred to as dynamic effects, which stem from either geometrical constraints or host-guest interactions, that alter energy barriers by impacting the formation of activated complexes. As a consequence of the combination of those dynamic effects, species inside nanopores were shown to, for instance, experience pressures up to 5 orders of magnitude higher than those in the bulk.^{31–33} The Grubbs group has recently applied these concepts in a study of the dimerization of nitric oxide inside carbon nanopores³⁴ and found excellent agreement with experimental results. Inspired by this body of work, many other researchers have taken advantage of molecular simulations to study confined environments. This section will highlight recent examples of studies that systematically decoupled the two main factors that contribute to such modified reactivity: host-guest interactions (*i.e.* electronic effects) and geometric constraints. We will compare selected studies and attempt to draw general conclusions about the interplay between mechanisms in spatially confined systems.

A range of selected simple reactions confined within two metal surfaces was studied by Balbuena and co-workers through the use of DFT calculations.³⁵ The dissociation of O₂ was simulated both on a free Pt surface and between two surfaces that were 4.9 Å apart. The authors found that the confined system provided more efficient charge transfer from the metal to the substrate, which decreased the energy gap between the bonding and antibonding orbitals of O₂. The larger population on the π^* orbital weakened the O–O bond, which was observed to be longer in the adsorbed molecule. This effect decreased the activation energy of the O₂ dissociation reaction by 35% when compared to the reaction on a single metal surface. BEP (Brønsted–Evans–Polanyi) relationships were also calculated for the dissociation of O₂, CO, and NO on different metal surfaces, and revealed a good correlation between adsorption and activation energies for all systems, with a negative relationship between the strength of adsorption and activation energy. The effect of confinement on activation energy decreased in the order CO > NO > O₂. The confinement effect on the activation energy of dissociation was found to be relatively independent of the metal, but highly dependent on the arrangement of the substrate molecules inside the confined space (Fig. 1). Molecules adsorbed in a conformation that optimized the interactions with both metal layers benefited from more charge transfer, which resulted in more effective bond weakening and, consequently, a larger decrease in activation energy. In other words, geometrical constraints that led to stronger host-guest interactions were key for the enhancement of reactivity in this study.

Another compelling study was carried out by the Huang and Fang groups, who used super-localization-based imaging

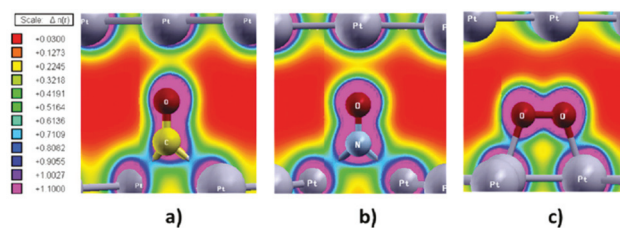


Fig. 1 Electronic density maps for molecules adsorbed between two Pt (111) surfaces at a surface–surface separation distance $H = 4.9$ Å. (a) CO, (b) NO, and (c) O₂. Reprinted with permission from ref. 35. Copyright 2011 American Chemical Society.

techniques to observe single molecules and study nanoconfined dynamic processes.¹⁷ The catalytic system consisted of Pt nanoparticles sandwiched between a dense silica core and a mesoporous silica shell. The pores were cylindrical (100 nm in length and 2.3 nm in diameter) and perpendicular to the core, having the top open to the bulk reaction medium and the bottom blocked by Pt nanoparticles. The oxidation of amplex red to resorufin (a fluorogenic reaction) was studied as a probe, and the authors found enhanced reaction rates in the confined reaction. The molecules had their rate of diffusion reduced by 5 orders of magnitude when confined inside the pores of the catalyst compared to the diffusion rate in the bulk, which came from the comparable hydrodynamic size of the product (0.51 nm) to the pore, the increased viscosity of the confined solution (4 times larger than in the bulk), and the interaction of the molecules with the polar siloxane groups on the silica walls. Such slow diffusion caused extremely high concentrations of molecules inside the pores, explaining the higher reaction rates. The same study was carried out using shorter channels (80 nm) and provided unchanged diffusion coefficients and adsorption–desorption constants, but a reduced rate constant. The diffusion coefficients and adsorption–desorption constants resulted in high effective concentrations inside both longer and shorter pores, but the longer ones were able to trap more reactants in the vicinity of Pt atoms, creating a larger reservoir of molecules ready to react, which translated into boosted rate constants. The Huang and Fang groups also published a follow-up study in the following year,³⁶ in which they studied the effect of the pore diameter by comparing smaller 2.2 nm and wider 3.3 nm pores. In addition to finding slower diffusion in the smaller ones, the authors demonstrated through fluorescence polarisation microscopy that substrate molecules inside the nanopores were prevented from π -stacking on the surface due to spatial restrictions, so the interaction with the reactant happened in an upright position, in which the long axis of the substrate was parallel to the length of the porous channels (Fig. 2). This created a favourable geometry for electron transfer, which reduced the activation energy of the reaction. Since the geometric constraints were more pronounced in smaller pores, the system with 2.2 nm pores provided higher catalytic activity. The work of the Huang and Fang groups present host-guest

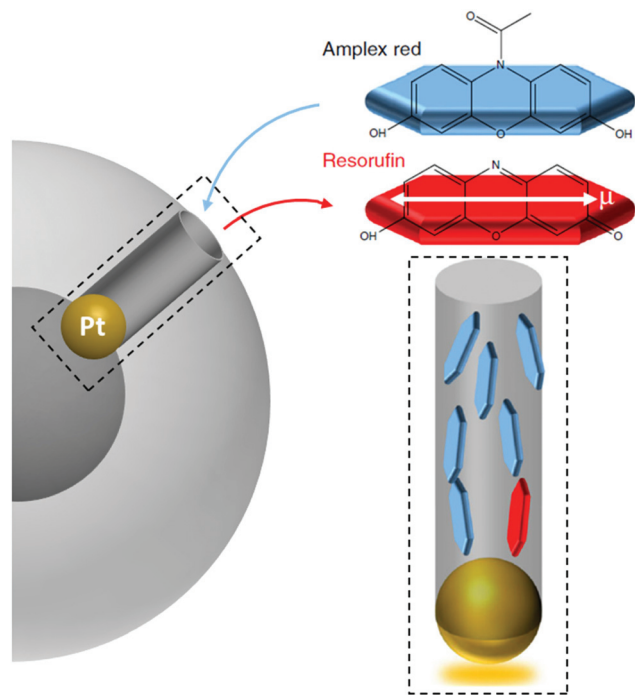


Fig. 2 On the right, amplex red oxidation to resorufin in silica pores catalyzed by Pt NPs. On the left, the view of a single pore and the geometrical constraints when the substrate approaches the catalyst. Adapted from ref. 17 and reprinted with permission from Macmillan Publishers, ref. 36, 2019.

interactions and geometrical restrictions that were both beneficial for the reaction rate. The former contributed to the slower diffusion rate inside the pores, which led to higher concentrations in the vicinity of the catalyst; while the latter was responsible for a lower activation energy by locking the substrate in a position that facilitated more effective interactions with the catalyst.

Host–guest interactions were studied using an interesting approach by the Peruchena group, which used quantum mechanical electron density distribution studies to investigate the methylation of benzene with methanol inside two different zeolite structures: H-ZSM-5 and H-Beta.³⁷ H-ZSM-5 has its porous structure formed by 10 rings that can fit a sphere of maximum diameter 6.36 Å, while H-Beta has slightly larger pores, formed by 12 rings and that can fit a sphere of maximum diameter 6.68 Å. The authors approached the analysis by splitting the interactions between the zeolite host and molecular guests into two types: primary and secondary. Primary interactions were the strong ones and related to the reaction itself such as adsorption, co-adsorption, bond breaking, and bond formation. Secondary interactions, on the other hand, were the weaker interactions that happened between the substrate and the atoms on the zeolite walls not directly involved in the catalytic process. Such secondary interactions are consequences of spatial confinement, which the authors systematically demonstrated by quantifying their contribution to the stabilization of the system for each individual reaction

step. The zeolite with smaller pores consistently proved to provide stronger confinement contributions with, for instance, a maximum of about 40% of stabilization due to secondary interactions, hence confinement, in the benzene adsorption step, against 26% of confinement stabilization in the larger pore for the same process. Confinement also had a higher contribution to the stabilization of the transition state in the smaller cavity, but the larger cavity had a lower energy barrier for this step because it had more space to allow the optimal transition state geometry. The authors found that, from an electronic standpoint, tighter-fitting pores improve host–guest interactions and therefore the stabilization by confinement, but such small spaces might compromise the formation of optimal geometric arrangements that are required in reaction pathways. This demonstrated that a high level of geometric restriction is not always beneficial in confined catalysis, since the confined geometry might not be the preferred one for reactivity.

Another illuminating example is the work of Chattaraj and co-workers,³⁸ who studied the [4 + 2] cycloaddition of 1,3-butadiene and ethylene not inside porous structures, but when encapsulated within two different molecular cages. The first one, ExBox⁴⁺, was a semi-rigid cyclophane with a box-like shape, 9.42 Å wide and 14.86 Å long. The second cage, CB[7] (cucurbit[7]uril), was a macropolycyclic compound formed from the condensation of seven glycoluril units, with a circular interior cavity of diameter 7.3 Å (Fig. 3). The authors found that the rate decreased in ExBox⁴⁺ by 4 orders of magnitude relative to the unconfined reaction while the reaction in CB[7] increased by 2 orders of magnitude. Electron-density analysis showed that confinement inside ExBox⁴⁺ caused the reactants to be parallel to the pyridinium rings of the cage, preventing the optimal positioning of the molecules to form the transition state. On the other hand, the reactants inside CB[7] did

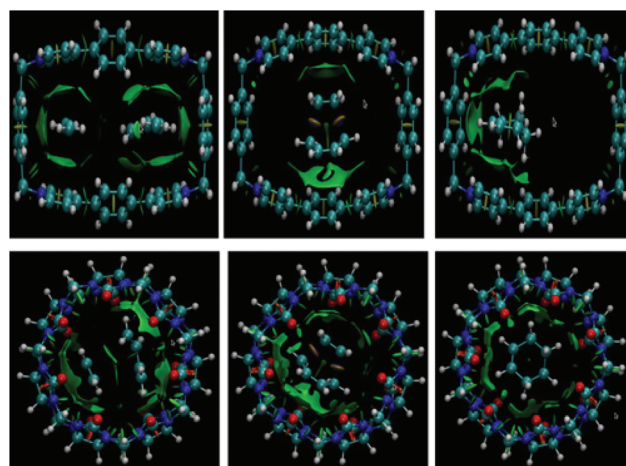


Fig. 3 Non-covalent interaction (NCI) surfaces. On top, left to right: reactant@ExBox⁴⁺, TS@ExBox⁴⁺, product@ExBox⁴⁺. On the bottom, left to right: reactant@CB[7], TS@CB[7], product@CB[7]. Reprinted with permission from ref. 38. Copyright 2015 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.

not interact with the cage but were instead held closer together, which facilitated the desired orbital overlap. This behaviour was seen as a consequence of a more favorable reactant-ExBox⁴⁺ than reactant-reactant interaction, while CB[7] had the opposite effect. Even though CB[7] increased reactivity in this specific system, it had previously been shown to hinder reactivity when used to encapsulate proteases in an earlier study. This example is the opposite of what was observed in Peruchena's work:³⁷ here, too effective host-guest interactions took place at the expense of the formation of the transition state, while spatial constraints favored interactions between the reactants and thus reactivity.

Carbon nanotubes (CNTs) are also a very useful confinement platform. An illustrative example is the DFT study of the dissociation of O₂ and N₂ on Pt NPs inside CNTs carried out by Bao and co-workers.¹⁸ The authors aimed to test whether reaction rates were affected by the nanotube diameter because of changes in curvature and distortion of the sp² hybridization on the carbon walls. It was found that, because of the strong interaction between metal clusters and carbon walls, the bonding energy between reactants and catalysts was weakened inside the nanotubes. For example, up to 40% reduction of the Pt-O₂ bonding energy was observed for Pt inside 0.95 nm tubes compared to an unconfined Pt-O₂ complex. Such weakened bonding energy favored the dissociation of activated species from the Pt surface, which increased the reaction rate for this reaction. However, while bonding energy was continuously reduced for smaller tubes, catalytic activity formed a volcano plot, with maximum activity in pores of around 1 nm in diameter. This occurred due to a change in the rate-limiting step as the substrate-catalyst bonding energy is progressively weakened. In tubes larger than 1 nm, where Pt-O₂ bonding energy was only slightly weakened, the rate of the final desorption step improved but remained limiting, and the reaction rate was enhanced. In tubes smaller than 1 nm, where Pt-O₂ bonding energy was too weak, the initial step of substrate adsorption to the catalyst was compromised and there was a steep drop in the reaction rate. This work, although not considering the effects of geometrical constraints explicitly, is unique in demonstrating the fine line between favourable and unfavourable effects of host-guest interactions on the rates of confined reactions.

The studies presented in this section were chosen as an attempt to showcase the richness and complexity of confined mechanisms. We have highlighted the role of host-guest interactions and geometric constraints in spatially confined catalytic systems, which are summarized in Table 1. Both effects

have proven to have the potential to be beneficial or detrimental to catalytic systems, depending on the specifics of each process. Geometric constraints, for instance, were shown to positively influence reaction rates by being able to bring reactants closer to catalysts and optimizing interactions, by forcing reactants closer together and facilitating orbital overlap, or even by promoting unlikely geometric arrangements that exposed reactive sites. Such geometric arrangements, however, can instead be destabilizing and lead to an increase in activation energy, as is the case of the methylation of benzene inside the pores of H-ZSM-5 zeolites. Host-guest interactions are no different and have also proven to be capable of significantly benefiting spatially confined systems by enhancing favourable charge transfer, contributing to slower diffusion and increased local concentrations of species in confinement, or by providing extra stabilization of transition states. Strong interactions might also compromise reactivity, for instance in the encapsulated [4 + 2] cycloaddition reaction, where stacking of the substrate to the host wall prevented the optimal transition state geometry. All things considered, confinement effects depend on different factors, and theoretical work such as those presented in this section can deconvolute these and uncover individual contributions. The complexity and peculiarity of such effects are evident, therefore there is the need for a greater understanding of these effects in order to use these principles to design more efficient and potentially greener catalytic systems.

3. Confinement in pores and cavities

Paolo Bettotti introduces his book on sub-micron porous materials by stating that “any material is porous if we consider that its crystallographic unit cells might act as cages for smaller molecular species”.⁴⁰ In fact, all the unique properties of pores are consequences of their sizes or – more precisely – their relative dimensions when compared to entities positioned within their interiors. Pores are openings, usually on solid structures, that increase the material's surface area and can come in various sizes, shapes, and degrees of uniformity. Pores are also sometimes referred to as channels or cavities depending on their morphology.⁴¹ Pores with diameters up to 100 nm, defined as nanopores,⁴² have dimensions where the interactions of the walls of the pore with atoms and molecules become significant, making them strong candidates for hosting confined chemical reactions.

Table 1 Effects of host-guest interactions and geometrical constraints in selected theory-based papers

Ref.	Catalyst	Reaction	Host-guest interactions	Geometrical constraints
35	Parallel Pt surfaces	O ₂ , CO, NO dissociation	Positive	Positive
17 and 36	SiO ₂ @Pt@m-SiO ₂	Amplex red oxidation	Positive	Positive
37	H-ZSM-5 and H-Beta	Benzene methylation	Positive	Negative
39	ExBox ⁴⁺ and CB[7]	[4 + 2] cycloaddition	Negative	Positive
18	Pt NPs in CNTs	O ₂ and N ₂ dissociation	Positive and negative	—

Zeolites are naturally occurring porous materials and represent the earliest and most widely studied cases of confinement.¹⁰ However, other types of porous materials, mostly mesoporous silica and porous carbon, have also been reported to present confinement effects that improve catalytic reactions. This section will review the role and activity of porous materials for catalytic applications that rely on confinement. At the end of the section, we will also review the use of nanostructured cavities to mimic the confinement effect observed in the cavities of enzymes.

3.1. Enhanced catalysis inside porous structures for green chemistry applications

The use of confined catalysis has now crossed the whole spectrum of catalytic processes and shown potential to improve chemical transformations across many facets of the chemical industry. We have here selected three main catalytic applications that significantly impact green chemistry: the sustainable production of high-energy-containing compounds for use as fuels, the greener manufacture of functionalized molecules to be used as building blocks or commodities, and the degradation of pollutants. Those three broad applications will be illustrated in this sub-section by selected examples of confined catalytic systems.

3.1.1. Confined catalysis for energy applications

Carbon dioxide conversion. Much of the energy consumed nowadays comes from fossil sources, which release CO₂ into the atmosphere and causes an unbalance of the natural carbon cycle. A solution that has been widely discussed is the capture of this carbon and its conversion to fuels, thus closing the anthropogenic carbon cycle.⁴³ Several carbon sequestration and conversion technologies are available in the literature,⁴⁴ and the use of confinement effects can greatly contribute to such catalytic transformations.

Copper has been historically used in the literature for CO₂ reduction because, in addition to being an abundant metal,⁴⁵ it can carry out both the CO₂ activation and *CO hydrogenation steps.⁴⁶ Nanoparticles of Cu₂O with cavities were used by Yang *et al.* to catalyze the electrochemical reduction of CO₂ into fuels.⁴⁷ Several systems have already been used to transform CO₂ into reduced C₁ compounds, but the formation of C₂₊ products remains a challenge. The challenge comes from the fact that Cu⁺, the species responsible for catalyzing C–C bond formation, is readily reduced at the negative potentials necessary for the electrochemical process. The nano-cavities in Cu₂O nanoparticles were able to concentrate carbon species (Fig. 4) and form a high carbon coverage on the catalyst's surface, which protected the Cu⁺ species from reduction and favored C–C coupling steps. A 9-fold enhancement of the C₂₊/C₁ product ratio was achieved when compared to the reaction on non-porous nanoparticles. Similar cavity-containing copper nanoparticles were used by Zhuang *et al.*, but to obtain C₃ alcohols from CO reduction.⁴⁸ Higher-order alcohols are desirable molecules to be used as fuels due to their high volumetric energy density, hence the interest in renewable routes for obtaining *n*-propanol. CO₂-to-CO reduction is well-established, but CO-to-

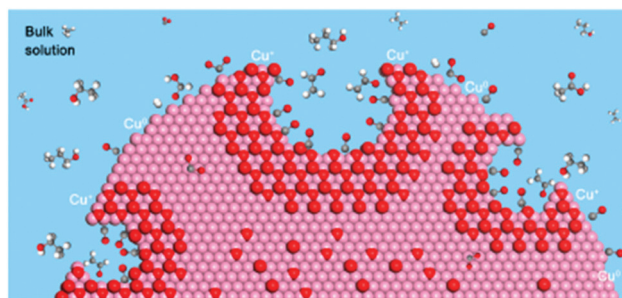


Fig. 4 Confinement of carbon intermediates in the nanocavities of Cu NPs, locally protecting the surface atoms from reduction. Reprinted with permission from ref. 47. Copyright 2020 American Chemical Society.

propanol conversions suffer from poor selectivity (around 10%). It was shown that a catalytic system with cavities could retain C₂ products in the vicinity of the catalyst surface and increase C–C coupling reactions, hence boosting C₃ selectivity. Finite-element simulations based on the angle of the opening of the cavities corroborated with the C₃/C₂ selectivity observed experimentally: small opening angles (<30°) afforded low conversions, large opening angles (>180°) gave high C₂ selectivity, and intermediate angles (45–90°) provided appreciable selectivity to C₃ products. While small angles limited the access of substrate to the catalytic surface and large angles favoured C₂ desorption, intermediate angles were able to both allow substrate adsorption and confine the C₂ intermediates close to the surfaces, leading to the formation of C₃ products. The formation of C₂ products on copper was also studied by the Tian & Liu groups, but using porous copper electrodes.⁴⁹ The authors found that methane is highly favoured in large 300 nm pores, while C₂ products were selectively produced in 30 nm pores. Comparing two electrodes with 30 nm-wide pores, but with different depths, high ethylene selectivity was found in short (40 nm deep) pores and high ethane selectivity in long (70 nm deep) ones. This difference was explained by the larger residence time of the substrate inside the longer pores, which allowed the formation of the fully saturated molecule. This fact was corroborated by the lower H₂ evolution observed in 70 nm-long pores. Copper electrodes for the reduction of CO₂ into C₂ products were also used by Zhang and co-workers but applying a novel halogen-assisted method to regenerate the electrode.⁵⁰ Iodide addition during the electrode reconstruction led to the formation of a hierarchical porous structure of copper nanowires, while bromide and chloride-assisted processes gave dense and quasi-cubic nanoparticles. All materials showed high electrochemical CO₂ reduction activity, but the porous structure afforded significantly higher selectivity to C₂ species. Such selectivity came from the accumulation of adsorbed CO* species on the surface of the porous electrode, which facilitated the *CO–*CO dimerization step and, consequently, increased the formation of multi-carbon species.

In addition to the direct reduction of CO₂, the use of confinement to assist the photocatalytic reforming of CO₂ with

CH₄ to produce syngas has been investigated.⁵¹ Mao *et al.* prepared Pt NPs confined within mesoporous CeO₂ nanorods to be used as catalysts for this process. The reformation of CO₂ with CH₄ is a highly endothermic reaction, which only happens at temperatures around 700 °C. In addition to the high energy consumption, such high temperatures lead to fast catalyst deactivation by the deposition of carbon on the active sites. It was found that confinement of the reactants and intermediates as the reaction unfolds created strong host-guest interactions, which activated oxygen atoms of CeO₂ that were at the interface with Pt NPs. These activated oxygen atoms facilitated all reaction steps, improving the catalytic activity of the system. The porous structure of the support also protected the active sites from carbon deposition and sintering, increasing the lifetime of the catalyst.

A significant outcome of CO₂ reduction technology is the sustainable production of methanol, a strong candidate as an energy vector⁵² that can be used, for instance, in direct methanol fuel cells. Datta *et al.* developed an efficient methanol electro-oxidation system by confining Pd NPs inside mesoporous carbon supports.⁵³ The authors compared fully confined, partially confined, and non-confined Pd NPs. The fully confined NPs were obtained through a silica-template procedure which assured that the totality of the metal NPs was found inside the pores; the second system was obtained by a regular immobilization strategy that had NPs both inside the pores and on the outer surface; the third system was commercial Pd/C catalyst (no pores). The fully confined system contained only half the amount of Pd that was loaded on the other supports, yet it provided the highest anodic peak current density ever reported for a Pd catalyst. Confinement also enhanced the stability of the catalyst over 300 electrocatalytic cycles. This superior activity was attributed to the protection of the Pd NPs against oxidation and leaching, as well as enhanced anodic potential due to greater electron density inside the pores.

Molecular hydrogen storage. Another important area of sustainable energy production is the use of molecular hydrogen as an energy carrier. H₂ has the highest energy density of a chemical energy vector and provides carbon-free combustion, but storage of H₂ is a significant challenge. Many research groups have investigated the storage of atomic hydrogen as chemical compounds that can release H₂ *in situ* upon catalytic activation.^{54,55} It has been found that this activation can be less energy-demanding upon confinement of the H₂ precursor in appropriately designed nanopores.

The Jongh group used confinement inside porous carbon structures to tackle the H₂ storage challenge.⁵⁶ LiNH₂ is a strong H₂ carrier candidate, but it has the drawback of only releasing H₂ at temperatures above 300 °C. Ammonia is another H₂ carrier option, but the current commercial catalyst for NH₃ conversion to H₂ is based on Ru, an expensive metal. The Jongh group approached the problem by confining LiNH₂ in porous carbon and converting it first to NH₃. This conversion was favoured under confinement and carried out at 125 °C. The *in situ* generated NH₃ molecules were then con-

verted to H₂ when the support was doped with Ni, with catalytic conversion rates comparable to the traditional Ru-catalysed process. Overall, confinement effects afforded a system that converted LiNH₂ to H₂ at reduced temperatures and using earth-abundant metal catalysis. The same group was also able to produce H₂ from ammonia using graphite nanoplatelets to embed nanoparticles of alkali metal amides (KNH₂ and NaNH₂) and transition metal catalysts (Ni and Ru).⁵⁷ Confinement increased the activity of the composites up to 4 times. The system with confined Ni at 8 wt% performed similarly to a 5 wt% state-of-art Ru/C catalyst (60–80% conversion) at 450 °C. The confined Ru-doped catalyst had significantly high activity (95%) at 375 °C, *versus* 475 °C needed for commercial Pd/C catalysts. The confined systems had some of the lowest activation energies for ammonia decomposition reported in the literature, which were attributed to geometrical constraints that forced the close proximity of negatively charged intermediates inside the pores.

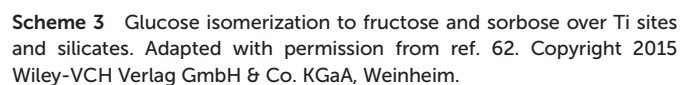
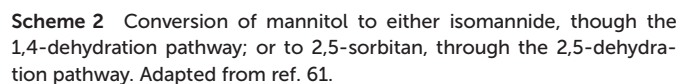
Looking at another H₂ carrier, Ding *et al.* investigated the decomposition of formic acid catalyzed by Pd NPs supported on carbon nanotubes.¹⁴ The authors produced and compared the activity of two versions of the catalyst: with metal NPs placed inside or outside the pores. The reactions were carried out at room temperature and, while metal-support interactions made both catalysts more active than pristine Pd NPs and Pd/C, the confined NPs afforded twice the TOF values of their unconfined counterparts. The confined system was also stable for 3 cycles, while the system with particles on the outside of the pores lost about 40% of their TOF upon recycling. Such deactivation was probably a result of CO poisoning, which was completely absent in the confined system.

3.1.2. Confined catalysis for commodity and specialty chemicals. Following the principles of green chemistry, important efforts have been geared towards the conversion of biomass into complex molecules to replace their sourcing from fossil resources. On the other end, the pursuit of more traditional petrochemical routes opens new avenues for chemical source diversification, for instance, *via* the gasification/Fisher Tropsch tandem reaction, now featured as a way for bio and urban waste recycling.^{58,59} These approaches come with synthetic challenges associated with the ability for these processes to be selective, a problem which confinement can help solve.

Biomass conversion reactions. Biomass is a renewable and abundant source of raw materials that can be transformed into high-value chemicals. These transformations proceed either by direct conversion, which takes advantage of the intrinsic oxygen content of bio-based compounds but requires the development of novel methods and infrastructure; or by full reduction to hydrocarbons or syngas followed by the reconstruction of functionality, which is redundant but utilizes the existing petrochemical infrastructure.⁶⁰ Many research groups have provided interesting examples of direct biomass conversion with improved selectivity in confined systems.

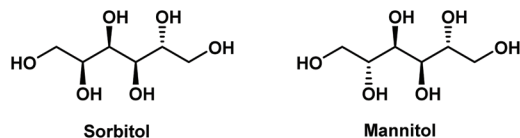
Yokoyama *et al.*, for instance, used DFT calculations to study the dehydration of mannitol, a product of the conversion

Another example is the work of Gounder and co-workers, who studied the conversion of glucose to sorbose, the precursor to synthetic ascorbic acid.⁶² The process can be catalyzed by Lewis acids, so the authors screened Ti-modified zeolites, mesoporous silica, and non-porous silica as catalysts. The mechanism (Scheme 3) involves the adsorption of glucose to the Ti site at the O5 position, followed by ring-opening to the open-chain form followed by coordination of the carbonyl in the C1 position to the Ti site. At that point, the intermediate can undergo either a 1,2-hydride shift, which gives the desired product; or a 1,5-hydride shift, which yields fructose. No reaction happened on non-porous and mesoporous (>1.50 nm diameter pores) supports, while catalytic activity was observed in microporous zeolite materials, showcasing the fundamental role of van der Waals stabilization of the intermediates in



narrow cavities. Zeolites with 12-membered ring pores (0.7 nm diameter) had fructose as their main product, while the ones with 8-membered ring pores (0.55 nm diameter) yielded primarily sorbose, suggesting that 1,5-hydrate shifts are unfavoured in the smaller pores. Pores formed with 8-membered rings (0.38 nm) were too small to fit the reactants and showed no activity.

In another study, the Borgna group¹⁵ contributed to the field with the conversion of cellobiose, a product of the hydro-



Scheme 4 Structures of sorbitol and mannitol, the sugar alcohols produced by the Borgna group¹⁵ from the hydrolysis of cellulose.

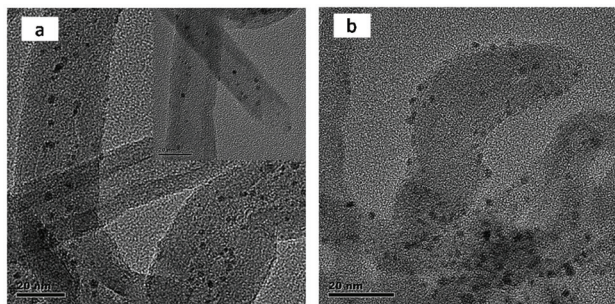
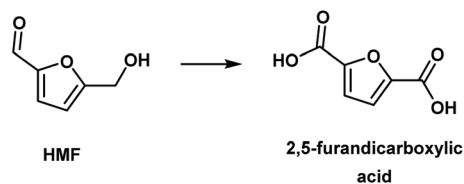


Fig. 5 TEM images of Ru NPs immobilized on CNTs, on the inner (a) or outer (b) surfaces. Adapted from ref. 15 with permission from the Royal Society of Chemistry.

lysis of cellulose. The authors produced sugar alcohols (sorbitol and mannitol, Scheme 4) over Ru nanoparticles immobilized on carbon nanotubes using closed-end and open-end nanotubes to assure the presence of nanoparticles inside or outside the structures (Fig. 5); the confined particles afforded higher catalytic activity. Three different tube diameters were studied (3–6 nm, 4–9 nm, and 6–15 nm), and higher activities were consistently found in smaller tubes. This higher activity was attributed to the prevention of leaching and lower electron densities on carbon nanotubes walls with higher curvature, corroborating with the strong electronic effects observed by Bao's molecular simulations.¹⁸

5-Hydroxymethylfurfural (HMF) is considered a central platform molecule in the biorefinery scheme, because it can be converted into many useful products.⁶³ The counterpart to this versatility is the need to develop effective and selective conversion pathways. Fang *et al.* looked at the aerobic conversion of HMF to 2,5-furandicarboxylic acid (Scheme 5), an important product in the pharmaceutical and polymer industries.⁶⁴ The authors designed their catalyst by incorporating a Co-containing zeolitic imidazolate framework ZIF-67 inside the pores of a mesoporous silica support. The composite was then calcined



Scheme 5 Conversion of 5-hydroxymethylfurfural (HMF) into 2,5-furandicarboxylic acid, as performed by Fang *et al.*⁶⁴

to yield ultra-thin cobalt oxide NPs confined inside the pores. The system afforded quantitative conversion and selectivity to the desired product; a 10-fold yield enhancement compared to wet-impregnated (unconfined) cobalt oxide nanoparticles on the same silica support. The confined catalytic system provided TOFs that were up to a factor of 150 times higher than those in the literature that were based on precious metals and applying harsher reaction conditions.

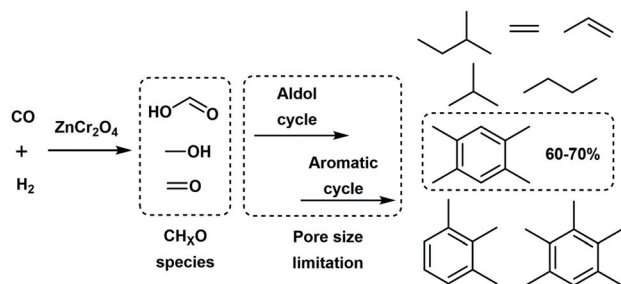
A large fraction of the world's biomass conversion is dedicated to the production of ethanol.⁶⁵ This compound, in addition to its wide utilization as a fuel, can also be converted into high-value products. Dai *et al.*, for instance, looked at the conversion of ethanol to butadiene, a monomer heavily used in the production of synthetic rubber.⁶⁶ They introduced different metals in H-Beta zeolites and found that a mixture of zinc and yttrium was optimal to turn the multi-step conversion into a one-pot reaction. The process inside the zeolite pores afforded higher butadiene selectivity and lower reaction temperatures when compared to the state-of-art methodology, which was attributed to the confinement of the species inside the micropores of zeolites. Similar systems with different pore sizes were tested for comparison: micro (0.8 nm, H-Beta zeolite), meso (2.8 nm, MCM-41), and macropores (24 nm, porous silica). When compared to the macroporous support, mesopores and micropores had a 50% and a 400% increase in the productivity of butadiene, respectively. The results were explained by a super-concentration of ethanol around the catalytic sites in tighter pores.

In all the examples delineated above, researchers highlighted how confinement can be used as a means to improve yields or help selectivity, which are two important challenges to tackle in the effort to achieve an efficient biorefinery.

Hydrocarbons and syngas chemistry. Hydrocarbon and syngas conversions are central processes in petrochemistry and as such, account for a large amount of existing infrastructure. They also offer an important green chemistry opportunity, as these processes play a key role in the developing industry of bio and urban waste valorisation.^{67,68} There are many contributions of confined catalysis to such conversions, some creative recent examples being highlighted below.

Arslan *et al.* gave one example of product selectivity in syngas conversion induced by confinement, in the context of the Fischer-Tropsch reaction.⁶⁹ The Fischer-Tropsch synthesis typically affords a broad oligomerisation product range. Unusually high selectivity of 70% in syngas conversion to tetramethyl benzene was observed using supported ZnCr_2O_4 on H-ZSM-5 zeolites. The metallic particles were responsible for the formation of C_1 oxygenated species, while the zeolite pores formed a “carbon pool” due to hindered rates of desorption and product diffusion. Aldol mechanisms then took place in the sub-nanoreactors allowing the chains to grow, but overgrowth was prevented by geometrical restrictions induced by the small zeolitic pores, which led to the high selectivity to small aromatics (Scheme 6).

Tan *et al.* also looked at syngas conversion but to prepare light olefins.⁷⁰ Like in the previous example, the authors uti-



Scheme 6 Simplified schematic of the Fischer–Tropsch synthesis performed by Arslan *et al.* using ZnCr_2O_4 supported on H-ZSM-5 zeolites.⁶⁹ ZnCr_2O_4 produced oxygen-containing intermediates, which grew thanks to the carbon pool created inside the pores. Hydrocarbons were formed through the aldol and aromatic cycles, but overgrowth was prevented by pore-size limitations. The system gave 60–70% selectivity to tetramethylbenzene in hydrocarbons.

lized Zn–Cr metal oxide particles as catalysts to convert syngas to methanol, and the acidic sites of silicoaluminophosphate (SAPO) zeolites to catalyze the dehydration of methanol to olefins. It was found that, while a physical mixture of the two active phases provided low light olefin selectivity and a handful of side products, a hierarchical core–shell catalyst was able to boost selectivity to light olefins. The core–shell structure led to methanol production occurring exclusively inside the pores, which ensured the effectiveness of the hydrogenation steps. The zeolite walls were able to adsorb water and the light olefin products, suppressing CO_2 formation by the water–gas shift reaction and preventing further hydrogenation of the olefins to paraffins.

Methanol can also be selectively converted into commodities.⁷¹ The methanol-to-olefin process is an example of a methanol conversion that can be efficiently improved by confinement effects, as in the work of Liu and co-workers.⁷² They designed a novel zeolite to enhance selectivity to propylene, a major building block molecule with several industrial applications. The conversion of methanol inside zeolites occurs through two mechanisms: the olefin cycle, which gives propylene and higher olefins; and the aromatic cycle, which goes through 5 or 6-membered ring intermediates to form ethylene. The latter is favored by a high concentration of acid sites, so the authors adjusted the Si/Al ratio in SAPO-14 to develop a zeolite with moderate acidity; such strategy disfavored the latter cycle and prevented the formation of ethylene. The aromatic cycle was also prevented by the ultra-small 8-ring cage on SAPO-14, which created a spatial restriction to the bulky intermediates. Within the olefin cycle itself, propylene was preferably formed over higher olefins because of the limited diffusion of C_{4+} molecules in the small openings. Propylene selectivity of 77.3% was achieved in the optimal system, the highest record of propylene selectivity in the methanol-to-olefin reaction. Another appealing outcome for methanol conversion is dimethyl carbonate, a platform chemical with extensive industrial applications. Zhao *et al.* used copper oxide nanoparticles supported on carbon nanotubes to accomplish

such a transformation.¹⁶ The authors used nanotubes with three different diameters (2–5, 5–10, and 20–50 nm) and found that, although all of them gave high selectivity to dimethyl carbonate, catalytic activity increased with decreasing pore diameter. This activity was attributed not only to the better dispersion of nanoparticles in the smaller pores but to the higher stability of reduced copper species, which was confirmed by XRD and XPS. In fact, both Cu and Cu_2O were the active species in the reaction and higher curvatures on carbon nanotubes increased the electron density on the inner surfaces, which was beneficial to the reduction of the oxides observed in the study. Those findings also perfectly corroborate with the findings of the Bao group.¹⁸

Confinement within pores has also played a key role in semi-hydrogenation reactions. Chai *et al.* constructed well-defined Ni centers coordinated to four framework oxygen atoms inside the micropores of chabazite.⁷³ In this unique design, the support worked as an inorganic ligand to the metal center. The catalyst was used in the semi-hydrogenation of acetylene, affording 100% conversion and 97% selectivity to the alkene and outperforming benchmark catalysts palladium–silver and the Lindlar catalyst. It was demonstrated that the electrostatic field created by the nano-scale reactor inside the zeolite was responsible for inducing the heterolytic cleavage of H_2 , which allowed the partial hydrogenation to happen so effectively. Another system was developed by the Gong group, who used SOD zeolites (pore size 0.36 nm) to confine Pd nanoparticles.⁷⁴ The authors performed acetylene hydrogenation and found 94.5% selectivity to ethylene, while Pd nanoparticles impregnated on the outer surfaces of the same zeolite afforded only 21.5% selectivity to the alkene under the same conditions. D_2 experiments with *in situ* DRIFTS and DFT calculations demonstrated that, while exposed Pd NPs allowed the substrate to directly adsorb on the metal and be fully hydrogenated, spatial restrictions made the pore-confined nanoparticles accessible only to the tiny H_2 molecules (0.28 nm). H_2 could be split inside the pores and hydrogen spillover was responsible for the formation of active OH species on the outer surfaces of the support. These OH groups were then approached by the larger acetylene molecules (0.38 nm), and a single hydrogenation step took place. Since the substrate did not strongly adsorb on the polar surface, it desorbed right after the first hydrogenation and led to high ethylene selectivity.

Selectivity in olefin metathesis is an important issue in the pharmaceutical industry that has also been addressed using confined catalysis. This reaction is very interesting for its high stereoselectivity, but it is difficult to selectively obtain macro (mono) cyclization products since this pathway competes with oligomerization. The Buchmeiser group performed olefin metathesis using the first-generation Grubbs–Hoveyda catalyst inside the pores of SBA-15 mesoporous silica.¹³ The authors reacted mesitylimidazole with an iodosilane, then the resulting salt was deprotonated with potassium hexamethyldisilazide (KHMDs) to give a free carbene. This free carbene was reacted with the first generation Grubbs–Hoveyda catalyst, giving it a

silane functionality that was used as the site of attachment to the mesoporous silica surface (Scheme 7).

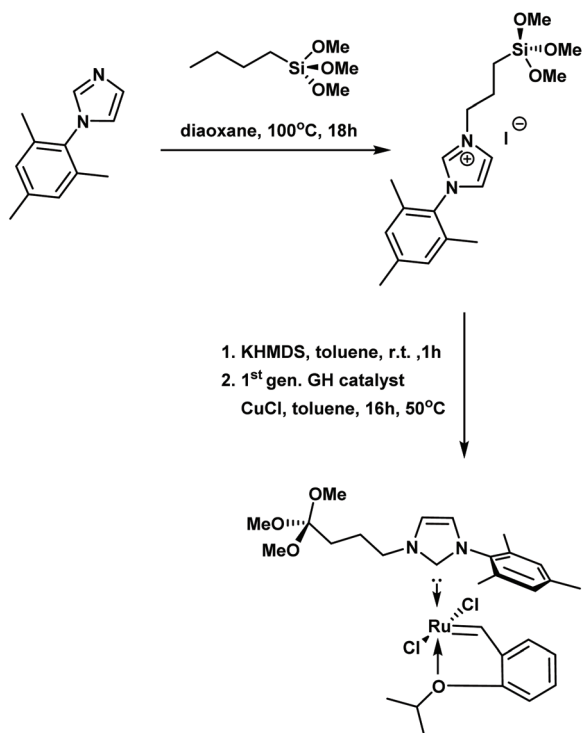
First, the surface of the support was modified to become non-polar and reduce the residence time of the carbonyl-containing substrates near the catalytic sites in the pores. This prevented ring-opening. Also, geometrical restrictions were used to limit the access of multiple substrate molecules to the catalyst at the same time, and limit oligomerization. Using surface-modified SBA-15 with 5.0 nm pores as a support, it was found that macro(mono)cyclization selectivity increased as the substrate hydrodynamic diameter increased from 0.75 to around 0.90 nm. However, larger than 0.90 nm substrates had a marked decrease in selectivity to the macro(mono)cyclic product. Such volcano plots are typical of confinement effects taking place with optimal selectivity for a specific type of substrate, resembling the confined mechanisms of enzymes.

Hydrocarbons are highly inert molecules, so introducing functionality selectively to these molecules is essential for their manipulation but experimentally quite complex. In this context, Paunović & Pérez-Ramírez performed a comprehensive study of methane halogenation in which several catalytic systems were evaluated.⁷⁵ The authors focused on chlorination and found that the reaction, which proceeds through a radical mechanism, was greatly affected by pore size when it occurred in the presence of zeolite catalysts. The catalytic activity also followed a volcano plot with maximum performance at a pore size of 0.5 nm, which is the sum of the kinetic diameters of

methane and chlorine radicals. Such activity and selectivity were attributed to the stabilization of radical intermediates and transition states by the tightly confined space.

Oxidized hydrocarbons, such as diols, can often represent important commodities or precursors to commodity chemicals. Monoethylene glycol is one of the most important diols in the chemical industry, but its production from ethylene oxide requires high temperatures (423–493 K) and a large excess of water ($\text{H}_2\text{O}/\text{ethylene oxide} = 20\text{--}25$) to prevent the self-condensation of the products and the formation of larger diols. Dai *et al.* approached this issue using Sn-zeolites.⁷⁶ The water-resistant Lewis acidic sites on the catalyst afforded high conversions, and supports with optimal pore sizes provided 99% conversion and 99% selectivity towards monoethylene glycol. It was found that the spatial restriction was responsible for limiting self-condensation, leading to highly increased selectivity to smaller molecules. The reaction temperature could be lowered to 313 K and the water content reduced to almost stoichiometric amounts, which directly helps reduce the energy needed to distill the final product. The researchers estimated a 50% reduction in the overall energy consumption of the process and demonstrated its applicability to a large variety of substrates. Another example highlighting oxidized hydrocarbons is the work of Liu *et al.* on the oxidation of toluene to benzaldehyde using copper complexes immobilized on mesoporous silica.⁷⁷ The authors found that the mesopores had the perfect geometry to hold the metals in a position to stabilize O_2 -bridged dicopper species (Fig. 6), and the conversion was carried out at room temperature.

These examples highlight the role that confinement can play in controlling the selectivity of complex, typically unselec-



Scheme 7 Functionalization of the 1st generation Grubbs–Hoveyda catalyst for immobilization on SBA-15 performed by the Buchmeiser group.¹³ PERMISSION?

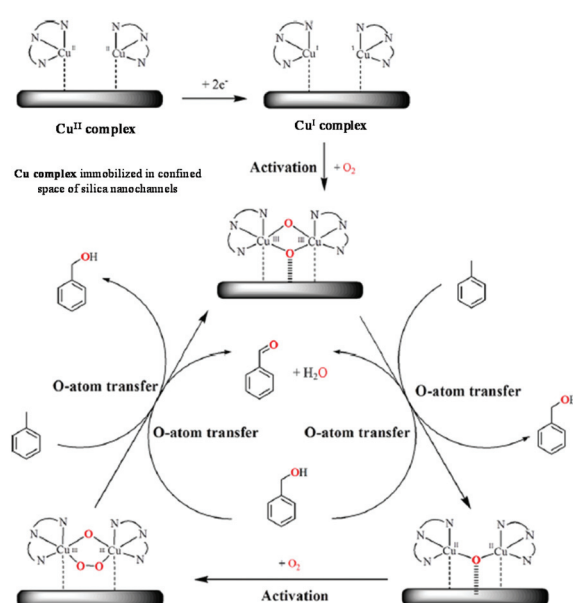


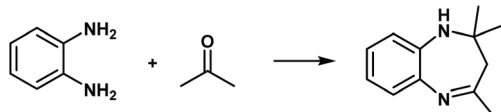
Fig. 6 Reaction scheme for the toluene oxidation mediated by copper bis(4-imidazolyl methyl)-benzylamine complexes immobilized in mesoporous silica nanoparticles. Modified from ref. 77. Copyright 2019 American Chemical Society.

tive processes by directly influencing either the speciation and diffusion of reactants or imposing geometrical constraints on the products formed.

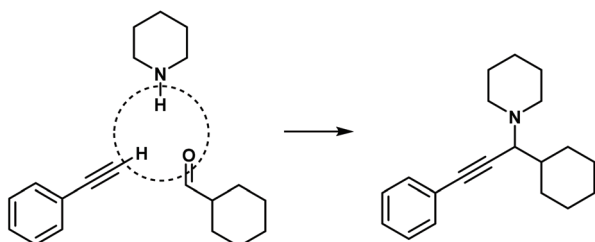
Fine chemicals. More complex and functional molecules are of special interest to the pharmaceutical and cosmetics industries and again call for the development of efficient and selective synthetic processes.⁷⁸ The focus is, in particular, on the reduction of precious metal loadings and improved selectivity and yields. Some fine chemical syntheses which have benefited from the use of confined catalysis are discussed below.

Godino-Ojer *et al.* studied the synthesis of benzodiazepines, which are nitrogen-containing heterocyclic compounds of great importance as pharmaceuticals.⁷⁹ Specifically, the authors looked at the reaction of *o*-phenyldiamine with different ketones catalyzed by sulfated zirconia (Scheme 8). Carbon xerogels (mesoporous) and microporous carbon were tested as supports, and the supported catalysts gave improved selectivity to the desired benzodiazepines. Catalysts supported on microporous carbon provided >90% conversions and 80% benzodiazepine selectivities with low catalyst load, which was attributed to both chemical and morphological properties.

Our group (Moore) has looked at the synthesis of propargylamines, another class of nitrogen heterocycles of relevance as medicines and chemical building blocks.¹² These molecules were made from the A^3 -coupling reaction, a one-pot process that combines an aldehyde, an amine and an alkene (Scheme 9). We used amine-functionalized mesoporous silica to immobilize monovalent copper species, which partially leached inside the pores of the MCM-41 support, creating an *in situ* catalyst delivery system. The limited diffusion inside the long channels caused a local increase in copper concentration,



Scheme 8 Synthesis of a benzodiazepine from *o*-phenyldiamine and acetone, one of the reactions performed by Godino-Ojer *et al.*⁷⁹ catalyzed by sulfated zirconia supported on mesoporous and microporous carbon materials.

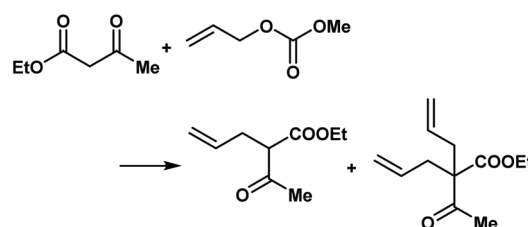


Scheme 9 The A^3 -coupling reaction (one-pot coupling of amine, aldehyde and alkene functionalities to give propargylamines) was performed by Terra *et al.*¹² with copper ions immobilized in the pores of functionalized mesoporous silica MCM-41.

which led our system to outperform the homogeneous state-of-the-art copper iodide catalyst. The local concentration inside the pores also allowed for the use of ppm levels of the metal catalyst, providing excellent yields of products with extremely low metal contamination. We built a substrate scope combining aldehydes like cyclohexanecarboxaldehyde and benzaldehyde; amines like piperidine, pyrrolidine and dimethylamine; and alkynes like phenylacetylene and 4-bromophenylacetylene.

Another interesting example is the work of the Tanaka group using Pd catalysts immobilized on a sophisticated multi-functionalized mesoporous silica support for C–C bond forming reactions that add allyl groups from allyl methyl carbonate and allyl alcohol to esters and phenols (Scheme 10),¹¹ a C–C bond forming reaction that gives crucial synthetic intermediates.

Using silane coupling reactions, the Pd complexes and tertiary amines were grafted on non-porous and mesoporous (1.6–5.3 nm) silica supports. A synergistic effect was observed, in which the nucleophile was activated by the tertiary amine *via* hydrogen abstraction, while Pd was responsible for activating the allylating agent. Non-porous structures showed little activity, while porous structures became more active at smaller pore sizes. TOF was improved from 220 on a non-porous support to 1240 when the reaction happened in 1.6 nm pores, even though the latter has only one-quarter of the Pd loading present in the former. The increase in activity at smaller pores was attributed to the spatial confinement of the active phases (Fig. 7). The activation of allyl alcohol by silanol groups was a



Scheme 10 Allylation reaction performed by the Tanaka group¹¹ using Pd catalysts supported on multifunctionalized mesoporous silica supports.

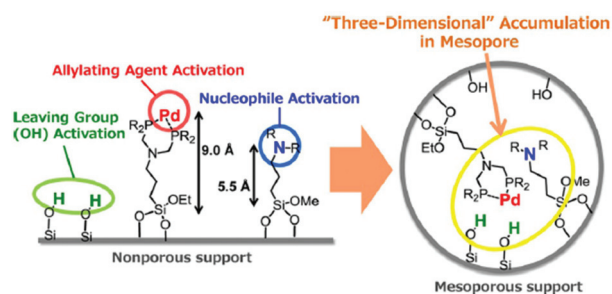


Fig. 7 Design concept used by the Tanaka group for confined allylation reactions. Reprinted with permission from ref. 11. Copyright 2017 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.

crucial improvement since allyl alcohols are simple and atom-economical but usually require activation by protic acid co-catalysts.

These examples highlight that even for targeted selective transformations, catalytic confinement can yield improvements in activity resulting in the reduced use of metals, hazardous co-catalysts, and promoters.

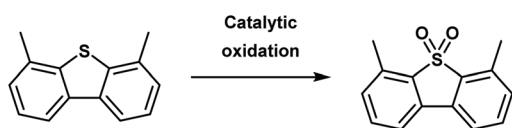
3.1.3. Confined catalysis for pollution mitigation. Even though the concepts of benign by design and pollution prevention have been advocated for by Paul Anastas and John Warner since the 1990s,⁸⁰ we are still far from having achieved the full transition to benign chemicals and pollution-free processes. Examples provided in the previous sections highlight methods by which catalytic confinement can make chemical products in a greener way. Another key area is the use of catalysis for pollution remediation, which can be improved in terms of specificity and efficiency through the incorporation of principles of catalytic confinement, as highlighted below.

One work by the Delyianni group addressed the oxidative desulfurization of 4,6-dimethyl-dibenzothiophene (Scheme 11),⁸¹ a common sulfur contaminant found in fuels. The oxidative desulfurization process is proposed as a more sustainable alternative to the high energy and hydrogen-demanding hydrodesulfurization process currently used. Acid-treated porous carbon materials were used as metal-free catalysts and were able to fully oxidize sulfides into sulfoxides or sulfones under mild conditions with hydrogen peroxide. The supports with the largest proportion of micropores performed better, which was attributed to confinement effects.

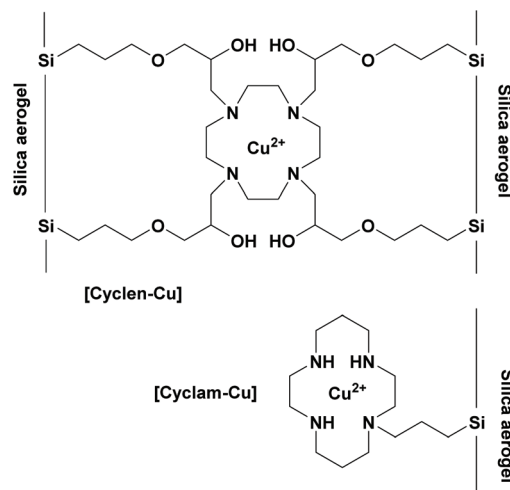
Another important class of pollutants is phenolic compounds, which are toxic and considered priority pollutants by the European Commission.⁸² Bereczki *et al.* approached the problem by anchoring cyclan and cyclen, which are tetraaza macrocyclic compounds, to the surface of silica aerogels, and coordinating Cu²⁺ ions to these ligands (Scheme 12) as catalysts for the oxidation of phenol in water.⁸³ The goal was to obtain solid composites for easy recovery and low metal leaching. The immobilized catalysts had TOFs up to 15 times higher than the free Cu-cyclan and Cu-cyclen, and this unexpected high activity was attributed to the confinement of the reactants and catalysts inside the mesoporous support.

3.2. Enzyme-like confinement

Enzymes are known for carrying out reactions with high selectivity and efficiency, in great part due to their structural features.⁸⁴ Enzymes are large molecules with sizes comparable to those of nanostructures, and they contain cavities that are



Scheme 11 Oxidation of 4,6-dimethyl-dibenzothiophene, performed by the Delyianni group using acid-treated porous carbon materials.⁸¹



Scheme 12 Structures of Cu-containing cyclen and cyclam anchored on silica aerogels, the catalysts used by Bereczki *et al.*⁸³ for phenol degradation.

induced by the folding of their constituent proteins. The cavities in enzymes can create confined reaction media with high specificity for selected solutes due to the orientation of complementary functional groups. Enzymes, therefore, represent an excellent model from which to develop active confined catalytic systems.

The Gooding group worked on the development of nanoparticles with cavities to mimic the three-dimensional structure of enzymes.⁸⁵ The authors synthesized oleylamine-capped Pt-Ni bimetallic nanoparticles consisting of Ni islands all over Pt spheres. The Ni islands could be etched using nitric acid to produce Pt nanospheres with cavities, where the hydrophobic ligand remained only on the external surface of the material; while the oleylamine could be removed by calcination at high temperatures (Fig. 8).

Different proportions of the two metals gave materials with either discrete cavities (PtNi₂), which the authors referred to as a nanozyme, or with interconnected pores (PtNi_{2.5}). The nanozyme consisted of a sphere of about 8 nm diameter, with cavities of approximately 1 nm diameter. These nanomaterials were tested as catalysts for the oxygen reduction reaction. Compared to the non-etched particles, TON was improved 1.6 times by the interconnected pores, 3 times by the surfactant-free particle with isolated cavities, and 3.6 times by the nanozyme. Although confinement effects were noticed in both pore-containing samples, discrete cavities performed better than interconnecting pores, which was attributed to the better isolation of nano-environments in the former, closer to the idea of a nanoreactor. The surfactant mimicked the effect of enzyme folding, selectively directing ionic structures away from the hydrophobic surface and into the hydrophilic cavities. In a follow-up study, the same group investigated the impact of channel dimensions on catalytic activity.⁸⁶ Three different cavity size distributions were explored, namely small (69% of the cavities below 2 nm), medium (52% below 2 nm),

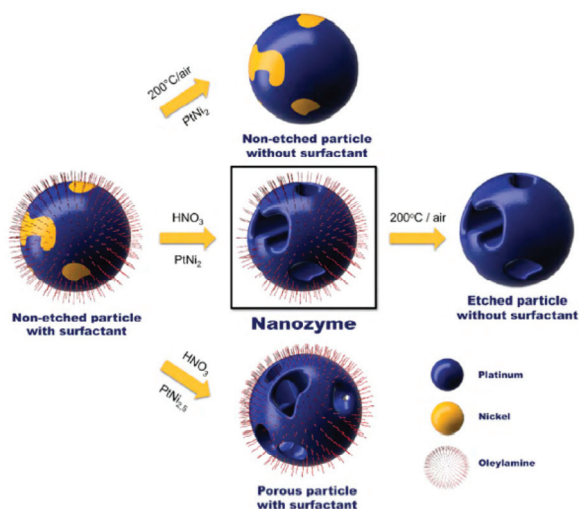


Fig. 8 Synthesis of nanozyme by acid-treatment of oleylamine-capped Pt–Ni NPs. Calcination was used to remove the surfactant and to increase Ni content led to interconnected pores. Reprinted with permission from ref. 85. Copyright 2018 American Chemical Society.

and large (34% below 2 nm). Small cavities were more active at low overpotentials, where the kinetics are dominated by electron transfer, related to the catalyst's intrinsic activity. Larger cavities gained more activity at higher overpotentials, when mass transfer had a higher influence. The authors performed physicochemical modeling and found that, because H^+ species could travel against a concentration gradient, they accumulated inside the cavities. Because of their dimensions, smaller cavities had larger concentrations of H^+ , which were significantly elevated compared to the bulk solution. On the other hand, the larger, neutral O_2 species did not concentrate. Hence, the high intrinsic activity of the smaller cavity catalysts could be attributed to the accumulation of H^+ , while the larger cavity systems dominated under higher potentials where O_2 diffusion into the cavity became rate-limiting.

This example highlights the possibility of engineering enzyme-like reaction spaces within rigid but multifunctional inorganic materials. It experimentally explores the factors that can be used to improve catalytic systems and again highlights the potential to borrow the concept of confinement from Nature and apply it to heterogeneous catalysis.

4. Confinement by encapsulation

Molecular confinement can also occur inside hollow materials. Those materials are defined as solid structures with space inside one or more distinct shells. Due to the void space within these materials, the shells can act as hosts and are capable of trapping guest molecules inside them, such as reactants and catalysts. Once the guest molecules are within the host, they are in a different nano environment and isolated from the bulk solution medium, being the truest examples of

nanoreactors.⁸⁷ Herein, we will define this approach as confinement by encapsulation.

Compared with porous hosts that are mostly restricted to silicon and carbon-based materials, encapsulation usually encompasses a wider range of hollow materials, allowing much richer chemistry. Different configurations can be used in the preparation of hollow materials for confinement by encapsulation. Gonzalo Prieto *et al.*, recently reviewed the main hollow material architectures and the advantages of using them as catalysts.¹⁹ The most common configurations of hollow materials can be summarized as being single shell (Fig. 9a), single shell embedded with nanoparticles (Fig. 9b), yolk-shell or core-shell (Fig. 9c), rattle-type (Fig. 9d), multi-shell (Fig. 9e) and nanoparticles within double-shell (Fig. 9f). These configurations enable the materials to have diverse chemical compositions. For example, the shell composition can be based on metals, metal oxides, alloys, MOFs, carbon nitrides, porphyrins, polymers, *etc.*; and they can have sizes in the micrometre or nanometre ranges. These tuneable nano environments can enhance the catalytic activity and the selectivity of reactions using similar principles to those in confinement within pores but with the potential for creating more diverse nanoreactor architectures. Such improvements in reaction efficiency are fundamental in the green chemistry context. In this section, we will highlight some examples of confinement by encapsulation in catalysis using these main hollow material configurations, which are relevant in the context of green and sustainable chemistry.

4.1. Single-shell hollow particle

Single-shell hollow particles are the simplest example of hollow materials. The reactants diffuse from the bulk solution through the pores of the shell and are confined in the available inner void space. Since there is no other active site, their outer and inner surfaces are responsible for the catalytic activity.

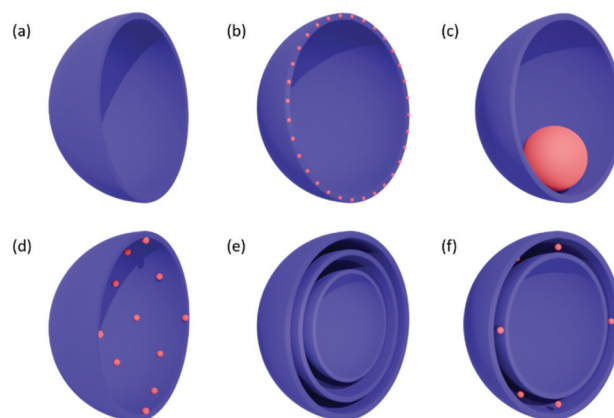


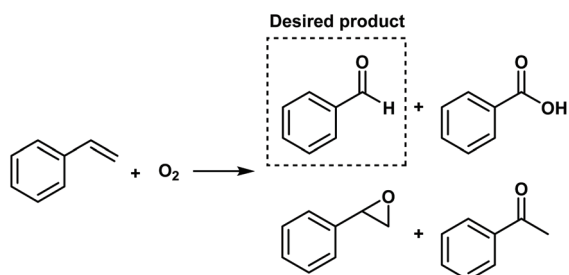
Fig. 9 Most common hollow material configurations: (a) single shell, (b) single shell embedded with nanoparticles, (c) yolk-shell or core-shell, (d) rattle-type, (e) multi-shell, and (f) nanoparticles within double-shell. Reprinted with permission from ref. 19. Copyright 2016 American Chemical Society.

This higher specific surface area, compared with their solid counterparts, can lead to higher reaction rates. For instance, our group (Moore) studied the catalytic activity of iron oxide nanoshells in the oxidation of styrene to benzaldehyde using molecular oxygen (Scheme 13).⁸⁸ The nanoshells gave significantly higher conversions than non-hollow NPs (84% *versus* 49–53%), despite a lower selectivity to benzaldehyde (65% *versus* 78–81%). The nanoshells were also etched, and the etched nanoshells gave higher styrene conversion (73%) and benzaldehyde selectivity (73%) than the non-etched NPs, providing evidence of the benefit of exposing the inner surfaces as reaction sites. The confined system was able to carry out the challenging oxidation of styrene using molecular oxygen, with high conversion and considerable selectivity to the partially oxidized product.

A study reported by Bin Qiao *et al.* demonstrated that PdCu single-shell nano boxes exhibit superior catalytic activity and reaction kinetics for the Cr(vi) reduction by HCOOH when compared with solid PdCu nanoparticles.⁸⁹ The higher activity was associated with the active sites available on the surfaces of the material that enhanced the atom utilization and the collision frequency through the confinement of reactants. Time-dependent UV-vis spectra measured at different temperatures showed lower activation energy of the reaction inside the PdCu nano boxes.

Rabia Jamil *et al.* also observed improved catalytic activity within single-shell hollow structures.⁹⁰ The performance of solid Pt–Ni NPs and hollow nano boxes were compared in the electrooxidation of methanol in fuel cells. Pt–Ni hollow nano boxes presented higher Pt% utilization and higher rate constants than the solid Pt–Ni NPs, which was confirmed by cyclic voltammetry measurements. It suggested that the inner surfaces of the hollow material participated in the reaction. While not explicitly explored, the rate enhancement may also have been influenced by a confinement effect within the nanovesels, which increased the supply of oxygenic species to the Ni and Pt sites and enhanced the activity of the catalyst.

In a study using Au nanocubes, Nazemi *et al.* were able to obtain high ammonia yields and faradaic efficiencies in the electrochemical reduction of nitrogen under ambient conditions.⁹¹ The authors compared the catalytic efficiency of the Au nanocages with solid Au nanoparticles of various shapes.



Scheme 13 Styrene oxidation and its possible products. The reaction was performed by Rak *et al.*⁸⁸ with high benzaldehyde selectivity inside hollow iron oxide nanoshells.

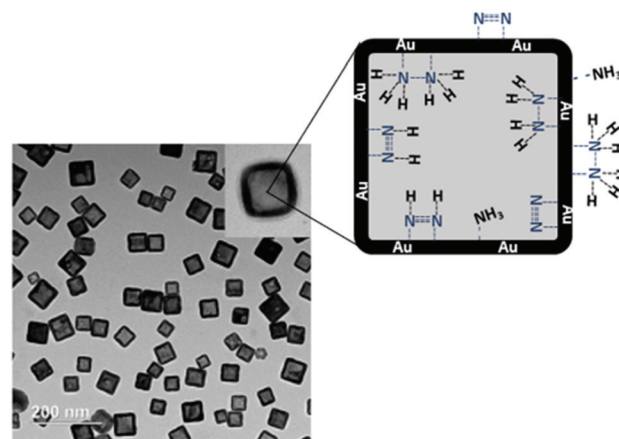
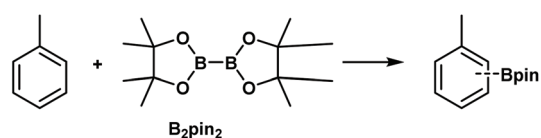


Fig. 10 Electrochemical synthesis of ammonia from N₂ and H₂O under ambient conditions using pore-size-controlled hollow gold nanocatalysts. Adapted with permission from ref. 91. Copyright 2018 American Chemical Society.

The Au nanocages (Fig. 10) gave higher activity, which was associated with the confinement of N₂ molecules within the cage. This confinement increased the frequency of collisions and the contact time of N₂ molecules with the catalyst, therefore facilitating the conversion of N₂ to NH₃.

Hollow nanocubes were also applied by the Du group in electrochemistry.⁹² The cages were template-made and the walls consisted of nanosheets of cobalt and molybdenum oxides and sulfides (CoMoO_x, CoMoS_x, and CoS_x). The high electrochemical surface area and charge transfer capability of the nanosheets were explored in water splitting reactions. The catalyst was used as the anode and commercial Pt/C as the cathode. The cell voltage required for the process of the reaction was 1.58 V, 32 mV less than the voltage required by the commercial IrO₂/Pt/C couple. XPS analysis demonstrated that the strong electronic interaction between the interfaces of the oxides and sulfides influenced the adsorption energy of oxygen-containing intermediates, which improved the catalytic performance and stability of the catalysts.

In another recent study, Li and coworkers explored the catalytic activity of a high-density atomic Pt nanobox in the direct C–H bond borylation of toluene (Scheme 14).⁹³ The nanoboxes achieved 99% conversion in 10 h at 120 °C, while the commercial Pt/C catalyst converted less than 70% under the same conditions. The Pt nanoboxes also presented better stability compared with the homogeneous catalysts. The improved performance was attributed to both the high-density of atomic Pt



Scheme 14 C–H bond borylation of toluene, as performed by Li and coworkers inside Pt nanoboxes.⁹³

species and the facilitated mass transfer and molecular diffusion within the hollow structures.

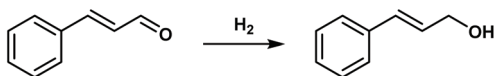
4.2. Single-shell embedded with nanoparticles

Pores or channels within the shell increase their surface area and allow the incorporation of nanoparticles inside them, forming hierarchical hollow materials (Fig. 9b). This configuration also enables a synergistic effect between the shell and embedded nanoparticles, contributing to the enhanced catalytic performance of the materials. Nanoparticles well-embedded in the pores are not as easily leached as from the external surface of a support, so this deactivation pathway is effectively suppressed in this type of architecture.

An illustrative study of the oxygen reduction reaction (ORR) with Fe_2O_3 nanoparticles dispersed in the mesoporous shell of an N-doped carbon hollow microsphere was published by Xiao *et al.*⁹⁴ The authors demonstrated that, compared with a series of hollow materials used as control samples, the Fe_2O_3 nanoparticles in the shell of N-doped carbon hollow microspheres exhibited higher activity and good electrochemical durability. This system provided a current density 18.5% higher than the commercial catalyst Pt/C. In addition, DFT calculations showed that the higher ORR activity was a result of the confinement in the microspheres, which lowered the reaction energy and facilitated the adsorption of O_2 molecules.

Another application for this architecture was the oxidative carbonylation of methanol to dimethyl carbonate, studied by Li and coworkers.⁹⁵ Among different catalysts based on carbon spheres and Cu nanoparticles, the one with Cu nanoparticles located in the mesopores of a carbon shell achieved the best catalytic activity. This improvement was associated with the better confinement of Cu nanoparticles and the high degree of Cu dispersion.

In a study using the cavities of MOFs, Kuang *et coll.* evaluated the performance of Pt catalysts in the selective hydrogenation of cinnamaldehyde to cinnamic alcohol (Scheme 15) under mild conditions.⁹⁶ The authors embedded the metal in two different matrices: a flexible one (Pt@ZIF-8), and a rigid one (Pt@ZIF-71). Carbon (20%)-matrixed platinum (Pt/C-20%) and PVP-stabilized Pt NPs (Pt-PVP) were also tested as reference catalysts. The hollow catalysts gave lower cinnamaldehyde conversions but higher cinnamic alcohol selectivities. Between the MOF-based catalysts, the flexible one exhibited higher catalytic activity and lower selectivity than the rigid structure. Although confinement decreased the diffusion of the substrate through Pt NPs, leading to lower conversions, it successfully inhibited the excessive hydrogenation of the substrate, leading to higher selectivities. The hypothesis was confirmed by con-



Scheme 15 Reduction of cinnamaldehyde to cinnamic alcohol, reaction performed by Kuang *et coll.* with high selectivities inside Pt-containing MOF structures.⁹⁶

structing additional MOF shells on top of Pt@ZIF-8 and Pt@ZIF-71 to confine exposed Pt NPs, which again provided lower conversions and improved selectivities.

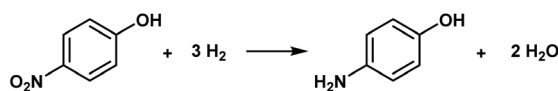
Choi & Oh also studied catalysts in MOF matrices.⁹⁷ They incorporated PdCo NPs in the structures of MOFs HZIF and HCS. The catalysts PdCo@HZIF and PdCo@HCS were applied in the reduction of 4-nitrophenol to 4-aminophenol (Scheme 16) and, compared with the non-hollow control materials, PdCo@HZIF and PdCo@HCS exhibited superior catalytic activities due to their unique structural features. The higher activities were attributed to facilitated mass transport through the hollow and porous shells and to the highly dispersed and confined incorporation of active bimetallic PdCo nanoparticles, which did not agglomerate and leach.

4.3. Yolk-shell and rattle-type hollow particles

Both yolk-shell and rattle-type configurations are hierarchical structures where the active site is encapsulated inside a hollow porous shell. The only difference between these configurations is the structure of the core: while the core of a yolk-shell architecture is a single solid particle (Fig. 9c), rattle-type is made from dispersed nanoparticles inside the shell (Fig. 9d). The encapsulation of the outer shell provides a confined nano environment where species can be well isolated and effectively stabilized. For instance, a recent study by Ma *et coll.* compared the activity of Ni NPs immobilized both inside (rattle-type encapsulation) and outside the hollow silica spheres.⁹⁸ The catalysts were used in the dry reforming of methane and the particles confined inside the hollow spheres gave higher conversions due to a higher dispersion of Ni particles and increased coke resistance.

Several other studies in the literature have also taken advantage of the higher concentration of species between nuclei and shells in yolk-shell structures. For example, the Xie group obtained high ethanol and acetaldehyde conversion and selectivity towards hydrogen production by low-temperature ethanol steam reforming, known as the LT-ESR reaction.⁹⁹ The catalyst used was a yolk-shell material composed of a CeO_2 nucleus and a Beta zeolite shell containing small Pt NPs on its interior surface, in a rattle-type fashion (Fig. 11). The limited diffusion inside the zeolite shells created an over-concentration of species in the interior of the catalyst, and both Pt and CeO_2 worked synergistically to improve H_2 selectivity and completely convert the substrates.

Another example is the work of Song *et al.*, which demonstrates the excellent catalytic activity of Pd-Fe NPs encapsulated in mesoporous silica (Pd-Fe@m-SiO_2) on the hydrogenation of a variety of olefins under ambient conditions.¹⁰⁰



Scheme 16 Reduction of 4-nitrophenol to 4-aminophenol, reaction performed by Choi & Oh inside PdCo-containing MOFs.⁹⁷

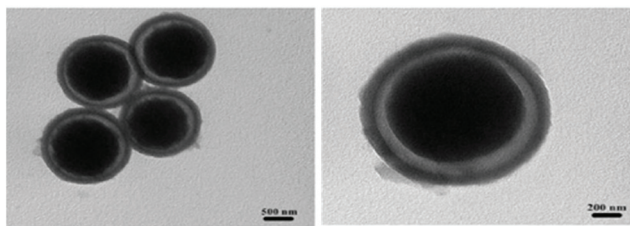


Fig. 11 TEM images of CeO₂@Pt-Beta at different magnifications. Reprinted with permission from ref. 99. Copyright 2019 John Wiley & Sons, Ltd.

Almost quantitative conversions of the olefins were obtained over only a few minutes (Table 2). The authors compared the activity of Pd-Fe@m-SiO₂ with commercial Pd/C and Pd nanoparticles loaded on the outside of Fe@m-SiO₂, denoted Fe@m-SiO₂/Pd. On the first four cycles, the catalytic activity of Pd-Fe@m-SiO₂ was nearly the same as that of Fe@m-SiO₂/Pd, but higher than that of commercial Pd/C. However, after 20 cycles, aggregation and detachment of Pd nanoparticles were observed for Fe@m-SiO₂/Pd and Pd/C, while the catalytic activity of Pd-Fe@m-SiO₂ was fully retained. The superior activity was attributed to the yolk-shell structure, which prevented the aggregation and deactivation of the catalyst and increased the local substrate concentration around the Pd NPs.

Cabello & Davoglio demonstrated that the confinement of reactants in a yolk-shell structure of MnO₂ hollow nanocages with encapsulated PtAu bimetallic nanoparticles reduced the

diffusion of intermediate species to the bulk solution.¹⁰¹ This led to a higher overall reaction efficiency for the oxygen reduction and glycerol electro-oxidation reactions.

The Quoc Cuong Do group investigated the effect of shell thickness and void size by comparing different yolk-shell structures of Fe₃O₄@SiO₂ in the degradation of acetaminophen, an emerging pollutant.¹⁰² It was demonstrated that materials with larger hollow void spaces and thinner silica shells performed better. This observation was attributed to the higher quantity of acetaminophen molecules that could be confined next to the Fe₃O₄ cores in larger voids. The Ren group reported a similar observation for the oxidative carbonylation of methanol to dimethyl carbonate.¹⁰³ It was found that larger void space in rattle-type Cu NPs on porous hollow carbon spheres structures led to higher activities compared to yolk-shell copper in hollow carbon spheres.

Another important feature of yolk-shell and rattle-type core-shell particles is the tunable composition of both the core and shell, allowing the effective integration of different components. These can promote synergistic interactions between the core and shell and enhance catalytic activity. Sanchez-Ballester *et al.* observed the existence of a synergistic effect for CO oxidation; the authors used a nanostructured polyaniline (PANI) matrix as a platform to create different yolk-shell compositions (Fig. 12).¹⁰⁴ The catalysts tested were: PANI nanocapsules loaded with Cu ions, Cu-yolk, Cu-yolk in Cu-shell, Cu-yolk in Au-shell, a physical mixture of Cu-yolks and Au-shells, and commercial Cu NPs. The hierarchical structure Cu-yolk in Au-shell gave the most effective catalyst, converting 82% of the substrate at 300 °C after 53 minutes. It was followed by the physical mixture of Cu-yolk and Au-shell (about one-third of the activity), then the pure Cu systems with lower activities. These results showed that both Cu and Au are important for the reaction to happen and that the confinement of Cu and Au together is responsible for the highest catalytic activity.

Another example has been demonstrated by Zhang *et al.*, in which a yolk-shell material with Pt-CeO₂ core and UiO-66-NH₂ MOF shell showed superior catalytic activity than Pt@UiO66-

Table 2 Scope of the hydrogenation of olefins with Pd-Fe@meso-SiO₂. Reaction conditions: Olefins (2.5 mmol), Pd-Fe@meso-SiO₂ (5.0 mg), ethanol (5 mL), 25 °C, under 1.0 MPa H₂ atmosphere. Data from ref. 100

<chem>c1ccccc1C=O</chem> $\xrightarrow[\text{H}_2, 25^\circ\text{C}]{\text{Pd-Fe@meso-SiO}_2}$ <chem>c1ccccc1CO</chem>					
Entry	Olefin	Product	Time (min)	Conv. ^a (%)	TOF (h ⁻¹)
1	<chem>c1ccccc1C=O</chem>	<chem>c1ccccc1CO</chem>	15	>99	7883
2 ^b	<chem>c1ccccc1C=O</chem>	<chem>c1ccccc1CO</chem>	15	0	—
3	<chem>CC1=CC=C(C=C1)C=O</chem>	<chem>CC1=CC=C(C=C1)CO</chem>	10	>99	11 824
4	<chem>COc1ccc(C=O)cc1</chem>	<chem>COc1ccc(CO)cc1</chem>	10	>99	11 824
5	<chem>c1ccc2cc(C=O)ccc2c1</chem>	<chem>c1ccc2cc(CO)ccc2c1</chem>	25	>99	4730
6	<chem>C1=CCCCC1</chem>	<chem>C1CCCCC1</chem>	40	>99	2956
7 ^c	<chem>ClC1=CC=C(C=C1)C=O</chem>	<chem>ClC1=CC=C(C=C1)CO</chem>	10	>99	11 824

^a Conversion was determined by GC, and the identity was ascertained by GC-MS. ^b 5.0 mg Fe@m-SiO₂ was adopted as a catalyst. ^c No dechlorination product was detected and the selectivity for 1-chloro-4-ethylbenzene.

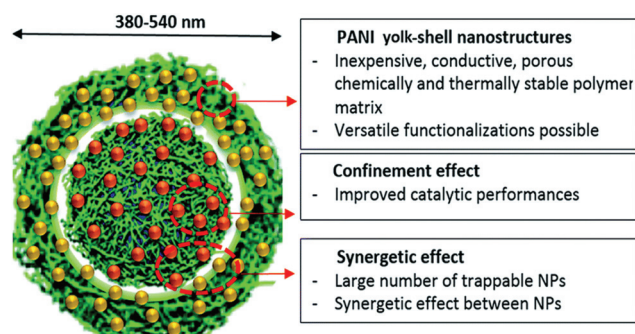
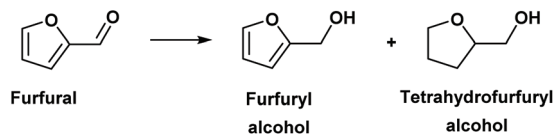


Fig. 12 Representation of the polyaniline nanostructure PANI, used by Sanchez-Ballester *et al.* to support Cu and Au NPs for confined CO oxidation reactions. Reproduced from ref. 104 with permission from The Royal Society of Chemistry.



Scheme 17 Furfural hydrogenation, the reaction performed by Zhang *et al.* in Pt/CeO₂@MOF catalysts.¹⁰⁵

NH₂ for the hydrogenation of furfural (Scheme 17).¹⁰⁵ The authors were able to confirm, by hydrogen temperature-programmed reduction, that Pt and CeO₂ possessed a strong interaction that modulated the d-band center and surface electron density of Pt, which led to the enhancement of the catalytic activity. The relativity towards the two hydrogenation products was dependent on the thickness of the MOF shell, having the thickest shell (30 nm) provided >99% selectivity towards furfuryl alcohol.

Furthermore, the tunable structure of the yolk-shell and rattle type configurations, *i.e.* space between the core and shell, as well as the thickness and pore size of the shell can influence the selectivity of reactions. The Tsubaki group observed that a core-shell structure of Zn-Cr oxide and zeolite SAPO-34 gave notably higher light olefin yields and lower CO₂ selectivity for the methanol-synthesis and methanol-to-olefin reactions that proceed as part of the Fischer-Tropsch process, compared with the physical mixture of both.¹⁰⁶ DFT calculations demonstrated that the SAPO-34 shells protected the Zn-Cr cores and limited the water gas shift (WGS) reaction as the adsorption energies of water are much higher on the SAPO-34 shell than on the Zn-Cr core. The yolk-shell structure also allowed better mass transfer and diffusion for a better synergistic effect. Furthermore, the confinement effect of the encapsulated catalyst inhibited side reactions and enhanced the light olefin selectivity (Fig. 13).

A final example selected with this architecture is a work of the Güttel group showing that cobalt-based core-shell catalysts (Co₃O₄@m-SiO₂) provided higher methane selectivity in the CO₂ hydrogenation reaction than the simple supported

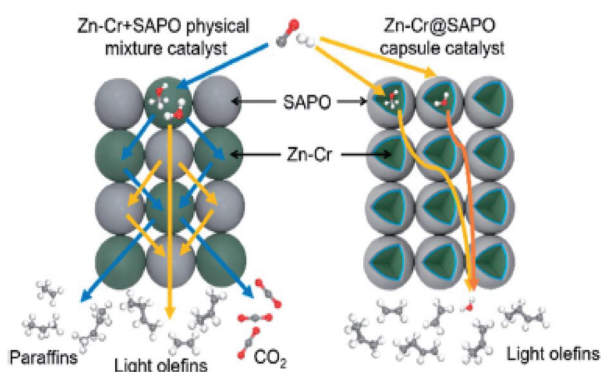


Fig. 13 The two different mass transfer routes over the Zn-Cr physical mixture catalyst and capsule catalyst. Reproduced with permission from ref. 106 with permission of the Royal Society of Chemistry.

system. Such higher activity was attributed to the confinement effect existing in the yolk-shell structure, which increased the probability for re-adsorption of CO on cobalt and favoured its methanation. However, for CO methanation, the core-shell catalyst was not efficient because the confinement of CO molecules improves the rate of C-O dissociation, causing deactivation by coking.¹⁰⁷ This highlights the necessity of understanding the molecular scale mechanistic processes to optimize the design of these catalysts for a given catalytic application, as demonstrated in section 1 of this review.

4.4. Double and multi-shell hollow particles

A hollow multi-shelled structure, as the name implies, is a material configuration with at least two shells of different sizes. This is similar to the structure of an onion, with an exterior shell protecting the interior ones (Fig. 9e). This configuration provides a large surface area with abundant reaction sites when the inner shells are catalytically active surfaces. Dan Wang and co-workers reported an interesting review of the correlation of the structure of hollow multi-shelled materials and their performance.¹⁰⁸ In the context of encapsulation, the isolated chambers in this architecture are able to provide additional free volume where confinement effects can also be observed. The multi-shelled structures also contribute to regulating mass transport, reducing effective diffusion distances for intermediate charges, ions, and molecules; and allow for multiple scattering of electromagnetic waves (Fig. 14).

An illustrative study was reported by the Yu group, comparing the performance of Bi₂WO₆ nanoparticles, single- and double-shelled Bi₂WO₆, broken microspheres and triple-shelled Bi₂O₃/Bi₂WO₆ hollow structures for the photodegradation of Rhodamine B.¹⁰⁹ Combined with H₂O₂, triple-shelled Bi₂O₃/Bi₂WO₆ (Fig. 15) had a 16-fold increase in activity when compared to Bi₂WO₆ nanoparticles in the degradation of the dye. This higher activity was attributed to the hierarchic structure enabling multiple light reflections and scattering patterns inside the material, a key advantage in this photocatalytic

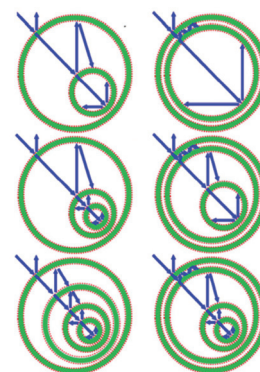


Fig. 14 Schematic of the multiple scattering of light by multi-shelled hollow materials. Extracted from hollow multishelled structures for promising applications: understanding the structure-performance correlation. Reprinted with permission from ref. 108. Copyright 2019 American Chemical Society.

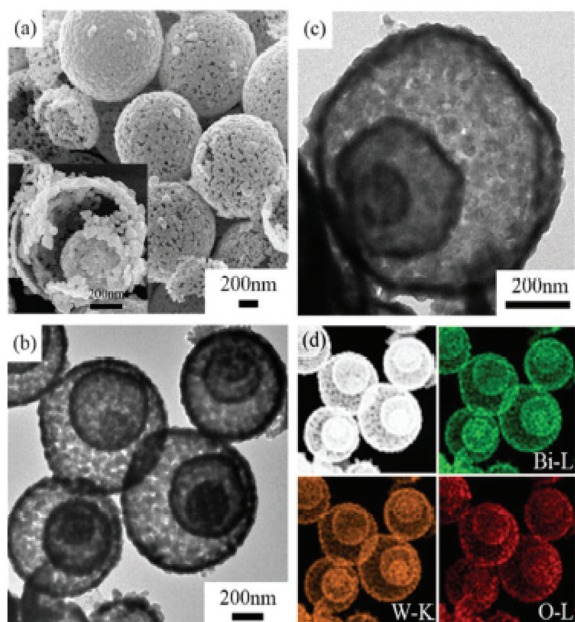


Fig. 15 (a) and (b) SEM and TEM images of BWO-MSH. (c) HRTEM image of the BWO-MSH. (d) HAADF-STEM element mapping. Reprinted from ref. 109 with permission from Elsevier.

process. The confined environment with an abundant presence of active sites also increased the probability of effective collisions.

The Zhang group recently reported the performance of hollow multi-shell CuO micro boxes for the electroreduction of CO_2 to ethylene.¹¹⁰ The CuO micro boxes presented higher catalytic activity than solid CuO particles, which were used as the control experiment. The higher electrocatalytic activity was associated with faster electron transfer processes and the increased adsorption of CO_2 molecules. The reaction was studied in different electrolytes (K_2SO_4 , KHCO_3 , and K_2HPO_4), and a higher faradaic efficiency was measured when the counter anion could not form a buffer (K_2SO_4) to neutralize the OH^- species that are generated near the catalyst/electrolyte interface, demonstrating the important role of these species. This high conversion efficiency of CO_2 to C_2H_4 was attributed to the confinement of the OH^- species within the CuO micro boxes. The higher OH^- concentration accelerated the CO_2 electroreduction pathways. Another relevant work was reported by Dong and co-workers, who evaluated the activity of Ce-Mn nanoparticles and a series of hollow spheres containing Ce and Mn with different numbers of shells for the selective catalytic reduction of NO to NH_3 .¹¹¹ Catalytic activity in these systems increased with the number of shells, with the triple-shelled hollow material providing the best performance, followed by the double-shelled, single-shelled, and then the nanoparticles not contained within a shell. The activity of the materials was correlated with their structural configuration. The hollow materials confined NO molecules inside them, increasing the probability of effective collisions between NO molecules and the active sites. A higher number of shells pro-

vided more active surface area, hence enhancing the catalytic activity.

Multi-shelled systems were also employed by the Peng group; the researchers used a LaNiO_3 double-shelled nanocage catalyst for CO oxidation and compared it to LaNiO_3 nanoparticles, pure NiO, and La_2O_3 .¹¹² The LaNiO_3 double-shelled nanocages presented superior catalytic activity at low temperatures and the highest reaction rate. The superior performance was mainly attributed to the structural configuration of the material. This configuration allowed the catalyst to act as a nanoreactor, and to store and confine the reactants inside it.

All these examples highlight that multi-shelled materials may be used as powerful platforms affording higher surface area, the ability to confine greater numbers of molecules, the ability to control diffusion and multiple scattering of light within these materials. These designs enabled increased catalytic outcomes.

4.5. Nanoparticles within double shells

Materials with this configuration present the same advantages as multi-shelled ones, but the presence of confined nanoparticles between the shells enables synergistic effects to occur, hence more opportunities to improve catalytic activity (Fig. 9f). Yin *et al.* studied the photocatalytic hydrogenation of nitroarenes by comparing the photocatalytic activity of Ag_2O and Ag nanoparticles supported on double-shell hollow SiO_2 spheres in various configurations: confined between the silica shells with space, confined between the silica shells without space, and supported on top of the outer shell (Fig. 16).¹¹³ The Ag_2O and Ag nanoparticles confined between the hollow SiO_2 shells with space achieved the highest yields and selectivity towards aniline in organic and aqueous media. Using molecular dynamics simulations, it was shown that the confined

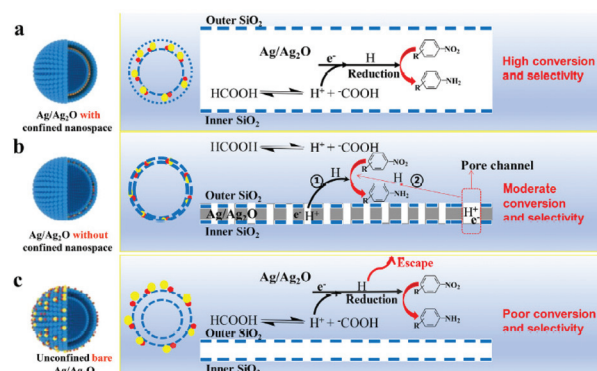


Fig. 16 Hydrogenation of nitroarenes. Hydrogenation model of photocatalysts with and without confined nanospace as well as unconfined bare $\text{Ag}/\text{Ag}_2\text{O}$. (a) $\text{SiO}_2\text{-Ag}/\text{Ag}_2\text{O}/\text{void}@/\text{SiO}_2$ catalysts with confined nanospace: high activity and selectivity are achieved in the reduction of nitroarenes under visible-light irradiation; (b) $\text{SiO}_2\text{-Ag}/\text{Ag}_2\text{O}@/\text{SiO}_2$ catalysts without confined nanospace: moderate activity and selectivity are obtained; (c) $\text{SiO}_2/\text{void}@/\text{SiO}_2\text{-Ag}/\text{Ag}_2\text{O}$ catalysts (unconfined bare $\text{Ag}/\text{Ag}_2\text{O}$): poor activity or selectivity are observed. Reprinted from ref. 113 with permission from Elsevier.

nano space significantly improved the collision probability of nitrobenzene and hydronium ions.

Zhao *et al.* studied the photodegradation of tetracycline within Ag nanoparticles confined in double-shell hollow TiO₂ photocatalysts as well as hollow TiO₂, Ag-decorated TiO₂, and Ag NPs confined within double-shelled TiO₂.¹¹⁴ The catalyst composed of confined Ag NPs achieved the degradation of tetracycline and excellent recyclability. The high catalytic activity was associated with the double-shell structure of the material and confinement effects. Electrochemical impedance spectroscopy (EIS) Nyquist plots and time-resolved transient photoluminescence showed that the confinement of Ag nanoparticles between the shells improved the lifetime of photo-generated charge carriers by inhibiting their recombination. Liang and co-workers reported an interesting study about the catalytic activity of hollow platinum nanocages that were encapsulated in porous carbon nitride for the hydrogenation of various nitroaromatics.¹¹⁵ They first tested the activity of this catalyst for the transfer hydrogenation of 4-nitrophenol (4-NP) and achieved >99% conversion and selectivity in only a few minutes. The material was further tested with another 23 nitroaromatics with different functional groups and completed the catalytic transfer hydrogenation for all of them within 10 min. The enhanced activity of the material was also attributed to its structural features. The material contained super thin walls with about six platinum atoms of thickness, hence high atom utilization efficiency and accelerated electron transfers. Furthermore, the synergistic effect between the iron oxides and Pt improved the catalytic activity.

Collectively these examples provide evidence for the ability to use hollow shells as effective catalytic systems for a variety of transformations of relevance to green chemistry, from targeted synthesis to the remediation of pollutants. Although confinement effects have been observed for all the presented structures, the heterogeneity of architectures and compositions of encapsulated systems poses a challenge in understanding and generalizing the mechanistic factors responsible for the improvement of catalytic activity. Future works will undoubtedly focus on rationalizing such effects towards the development and optimization of catalytically active hollow shell systems.

5. Confinement in self-assembled structures

The preceding discussion has focused on the confinement of reactants and catalysts within spaces that have been prepared through the use of covalent interactions leading to the formation of rigid pores, cavities, or encapsulated systems. An alternative approach involves the confinement of reactants and catalysts within “soft” self-assembled supramolecules or at liquid–liquid interfaces driven by non-covalent interactions. Micellar catalysis, first explored in the late 1950s,¹¹⁶ is the most widely studied example of this form of confinement. Interest in micellar catalysis has accelerated in the past decade

due to the desire to replace environmentally hazardous organic solvents with water.^{117,118} Other related areas that will be discussed here include the use of liquid–liquid interfaces and the hydrophobic effect to accelerate reactions through on-water catalysis^{119,120} as well as the use of inherently nano-structured fluids such as ionic liquids (ILs) and liquid crystals to affect reactivity.^{121–123}

The supramolecular chemistry of organometallic compounds has also been largely explored in the context of confined catalysis, and several approaches can be found in the literature for building those systems. Strategically selected organic structures can simultaneously anchor metal centres and create confined environments where reactions take place. Moreover, metal–ligand coordination chemistry is particularly appealing because it allows very precise geometric control of atomic arrangements, hence greater knowledge of the reaction sites. Cyclodextrins^{124,125} and porphyrins,^{126,127} for instance, have been used to covalently bind metals; while molecular cages have been used to host metal ions by non-covalent interactions.^{128,129} Such structures can be also referred to as molecular barrels due to their large internal cavities.^{130,131} Excellent reviews on confined catalysis in organometallic structures can be found in the literature, including the work published by Reek and co-workers,²¹ and more recently by Sinha & Mukherjee.²²

Non-covalent self-assembly as a method of confinement presents some inherent advantages from a green chemistry perspective. The complete architecture used for confinement does not need to be synthesized in its entirety which markedly reduces the synthetic complexity and the associated waste and energy requirements. The dynamic nature of self-assembled processes reduces diffusion limitations that are present in the case of more rigid pores and cavities.

However, as the confinement is driven by non-covalent interactions there is the need to ensure the desired architectures are maintained in the presence of additives such as reactants, catalysts, and solvents. The lack of rigid containment and the dependence on partitioning between phases on the confinement can also affect the substrate and reaction scope of these processes. These factors will be explored for selected representative examples that will be discussed in the following sections.

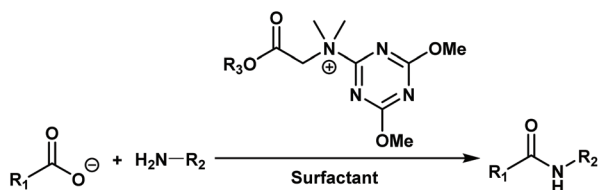
5.1. Micellar catalysis

One of the original embodiments of the concept of nanoconfinement to achieve chemical transformations has been through the use of micellar catalysis. Micellar catalysis broadly refers to the use of aqueous association colloids, typically micelles, to achieve improvements in reaction rates.¹³² The concept of confinement is central to micellar catalysis and the space in which the reactants are confined significantly affects their reactivity.¹³³ In micellar catalysis, reactants partition between the aqueous and the micellar pseudophase. The micellar pseudophase consists of both the hydrophobic micellar interior and the more polar, often ionic, micellar interface. Reaction kinetics in micellar systems depend upon the

effective local concentrations of reacting species within each pseudophase as well as the influence of the local solvation environment.¹³⁴

The partitioning of species between micellar and aqueous pseudophases impacts the kinetics of reactions performed in the presence of micelles. Unimolecular reactions are primarily affected by the local solvation environment which differs due to the polarity difference between the micellar and aqueous pseudophases.¹³⁴ Bimolecular reactions are affected by both the local solvation environment and the ability of reactants to undergo successful collisions. Consequently, bimolecular reactions involving ions are strongly affected by the nature of the surfactant. Ionic surfactants of similar charge to the reacting ion will decrease the extent of its partitioning into the micellar pseudophase whereas oppositely charged ions will preferentially adsorb. This effect is of particular importance for acid and base-promoted reactions since OH^- will be more closely associated with cationic surfactants and H_3O^+ to anionic surfactants.^{117,133,135} For hydrophobic compounds and ions attracted to the micellar pseudophase, there is the opportunity to generate rate enhancements of several orders of magnitude, from the substantial increase in the effective concentration of reactants in the micellar pseudophase relative to the overall solution.¹³²

A clear illustration of the bimolecular rate enhancement effect has been observed for the dehydrocondensation reactions between amines and carboxylates explored by Kunishima *et al.*, depicted in Scheme 18.^{136,137} Rates nearly 3 orders of magnitude greater were achieved with *N*-butylamine when sodium undecanoate rather than sodium propanoate was used in the presence of an amphiphilic dehydrocondensing agent. This was attributed to the increased proximity of reactants and the coupling agent due to the formation of micelles.¹³⁶ The addition of anionic or nonionic surfactants facilitated rate enhancements for non-amphiphilic carboxylates, although the magnitude of these rate increases was less than those for the amphiphilic carboxylates.¹³⁷ This can largely be attributed to the “dilution” of the carboxylate within the micelle by the added surfactant as well as the reduced propensity of the shorter chain carboxylates to partition within the micellar pseudophase. The use of cationic surfactants suppressed reactivity, which was attributed by the authors to increased rates of hydrolysis of the coupling agent. However, this may also have arisen from the repulsion of the cationic coupling agent by the positively charged micelles.



Scheme 18 Dehydrocondensation reaction between amines and carboxylates explored by Kunishima *et al.*^{136,137}

Ester hydrolysis can also be significantly influenced by the use of micelles due to the ability of these reactions to be acid or base-catalyzed. Tehrani-Bagha and Holmberg found that the rate of hydrolysis of ester-containing ammonium surfactants in the presence of a pH 7.5 phosphate buffer increased with the charge density of the headgroup.¹³⁸ Short-chain gemini surfactants (Fig. 17) led to rates of hydrolysis nearly an order of magnitude larger than structurally similar monomeric surfactants. Conversely, the rates of hydrolysis for anionic surfactants were found to decrease in the presence of cationic surfactants under basic conditions.¹³⁹ This was attributed to the competitive displacement of the OH^- ions by the substrate. Similar competition can also be observed by the addition of unreactive salts, with these ions displacing OH^- or H_3O^+ ions adsorbed to the micelle.¹⁴⁰ Similar general trends can be observed for acid-catalyzed hydrolysis where rates tend to increase for anionic surfactants and decrease in the presence of unreactive cations.^{140–142} Collectively, these outcomes emphasize the ability of micellar systems to achieve rate enhancements by increasing the association of substrates and catalysts but highlight the need for careful consideration of the co-location of these reacting species within the micellar pseudophase.

The hydrolysis of amphiphilic esters has even been proposed as a model of prebiotic chemistry.¹⁴³ The hydrolysis of ethyl caprylate occurs 900 times faster in the presence of sodium caprylate, the product of the hydrolysis. Initially presented as evidence for the self-replication of micelles, it was later discovered that the kinetics could be explained by phase transfer rather than micellar catalysis.¹⁴⁴ Nonetheless, the micelles formed were able to produce vesicles on acidification which could be induced by exposure to gaseous CO_2 . This illustrated the formation of complex self-assembled structures from a simple molecular precursor under ambient environmental conditions. This and related studies have led to the investigation of micelles as biomimetic structures for the design of reactive chemical spaces which will be discussed below.

5.1.1. Micellar catalysis as an enzyme mimetic. The ability to concentrate reactants from the bulk solution into a separate reaction space has drawn many parallels with the operation of enzymes. The ‘binding’ of reactants within the micellar pseu-

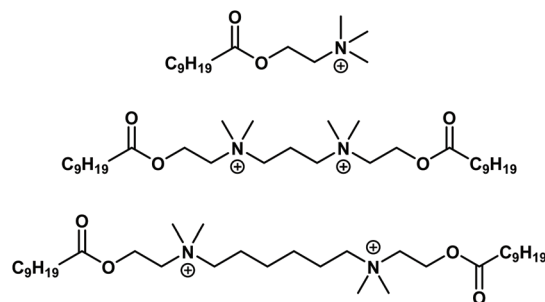
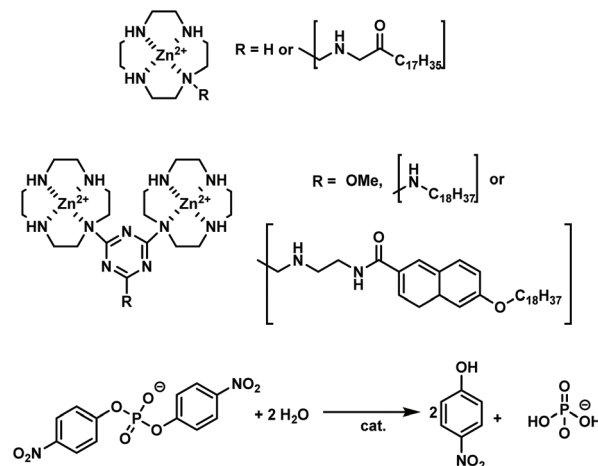


Fig. 17 Ester-containing gemini and monomeric surfactants used as substrates for hydrolysis investigations by Tehrani-Bagha and Holmberg.¹³⁸

dophase can be combined with appropriate functionalization of the surfactant headgroup to provide the intermolecular stabilization of transition states required to achieve improved reaction rates. These considerations need to be balanced with the ability of the surfactants to form aggregates and the partitioning of the reactants into the micellar rather than aqueous pseudophase but offer the potential to create systems akin to synthetic enzymes.

Nothling *et al.* recently expanded the ability of surfactants to act as enzyme-mimetics.¹⁴⁵ They prepared a novel surfactant-containing imidazole, carboxylic acid, and hydroxyl groups as an artificial catalytic triad inspired by the active site of α -chymotrypsin. In addition, cetyltrimethylammonium bromide (CTAB) and hexadecylguanidinium chloride co-surfactants were used to mimic respectively the hydrophobic binding pocket and the oxyanion hole typical of enzymes, and facilitate increased hydrogen-bonding interactions within the transition states. This combination afforded a catalytic system with greater activity for the hydrolysis of *p*-nitrophenol benzoate than the native enzyme as depicted in Fig. 18. Control experiments illustrated the need for all 3 components to be present for optimal activity. Central to the hydrolysis activity of this catalytic system was the confinement of both the novel surfactant and reactant within the micelle created by the guanidinium and CTAB surfactants. The previous example illustrates the use of nanoconfinement to achieve cooperative effects

between non-covalently bound catalytic sites. These cooperative effects have been explored by Gruber *et al.* who prepared amphiphilic zinc(II) 1,4,7,10-tetraazacyclododecane (cyclen) complexes for the hydrolysis of bis(4-nitrophenyl)phosphate, plasmid DNA, and a non-activated single-stranded oligonucleotide (Scheme 19).¹⁴⁶ It was identified that non-amphi-



Scheme 19 Zinc cyclen catalysts and the bis(4-nitrophenyl)phosphate hydrolysis investigated by Gruber *et al.*¹⁴⁶

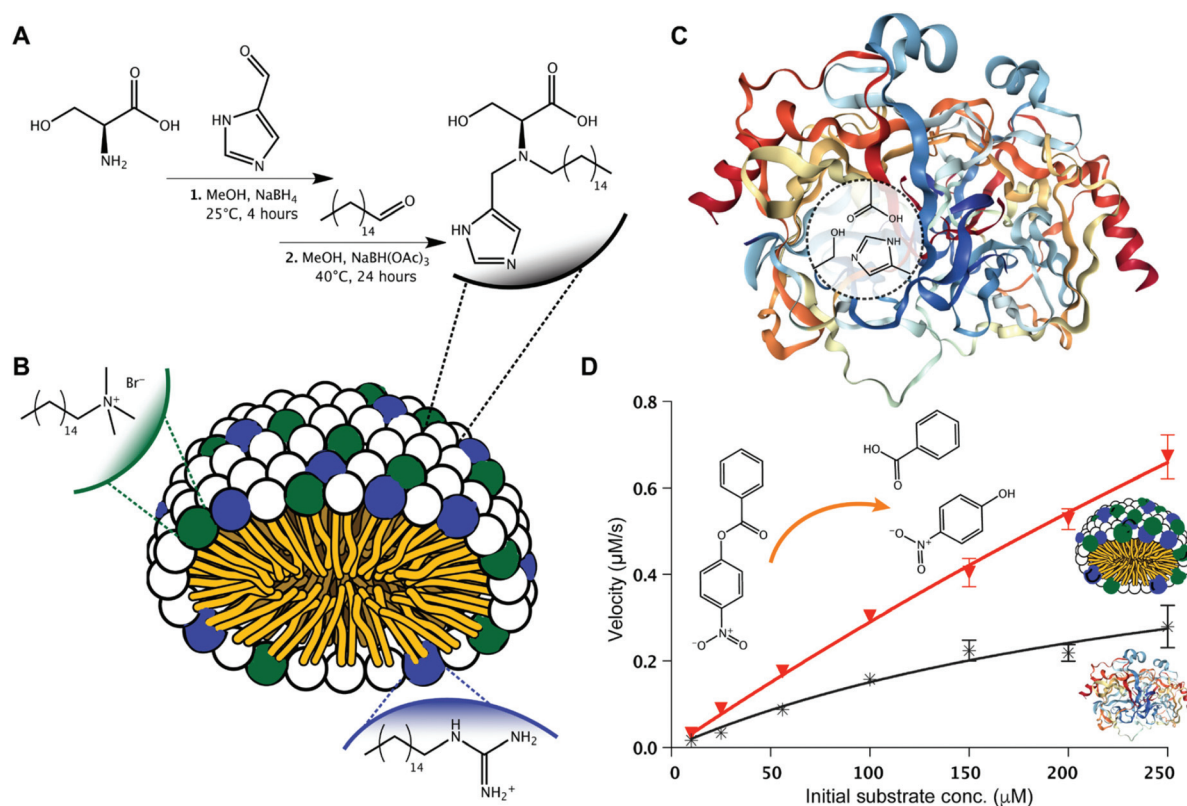


Fig. 18 Hydrolase-mimetic surfactant system investigated by Nothling *et al.* for the hydrolysis of *p*-nitrophenol esters. A) straightforward preparation of the surfactant; B) self-assembly of surfactant and co-surfactant; C) active site of a common hydrolase highlighting a hydrophobic pocket; D) hydrolysis of *p*-nitrophenol benzoate activity using the coassembly or the native enzyme. Reproduced from ref. 145 with permission.

philic cyclen zinc complexes were not catalytically active, whereas the amphiphilic complexes were active either as micelles or when combined with a phospholipid to form a vesicle. Structurally similar bimetallic zinc complexes had been previously used to highlight the importance of cooperativity between the zinc metal centres on the hydrolysis of phosphate diester bonds.¹⁴⁷ This illustrated that the most plausible explanation of the catalytic efficiency of the micellar zinc complexes is the cooperativity arising from the close proximity of metal centres within the micellar headgroups. Work by Solís Muñana *et al.* on related zinc 1,4,7-triazacyclononane (TACN) complexes demonstrated that micellization was critical for the transphosphorylation of 2-hydroxypropyl-4-nitrophenyl phosphate (Scheme 20).¹⁴⁸ In this case, there was an electrostatic attraction between the cationic surfactant and anionic substrate which had the dual role of increasing the local concentration of substrate and reducing the critical aggregation concentration of the surfactant leading to the substrate templating the formation of the catalytically active aggregates. The ability of the zinc complexes to form aggregates was closely associated with their resultant activity with non-amphiphilic TACN complexes resulting in initial rates over 1000 times less than the corresponding C₁₈ variants used above their critical aggregation concentration.

The above examples illustrate that the closely associated nature of surfactant headgroups, counterions, and adsorbed substrates provides an excellent opportunity to develop cooperative biomimetic catalytic systems. By selectively introducing functionalities to stabilize key transition states or pre-organizing multiple catalytically active moieties, more active and selective catalysts will be able to be designed, leveraging the concept of nanoconfinement.

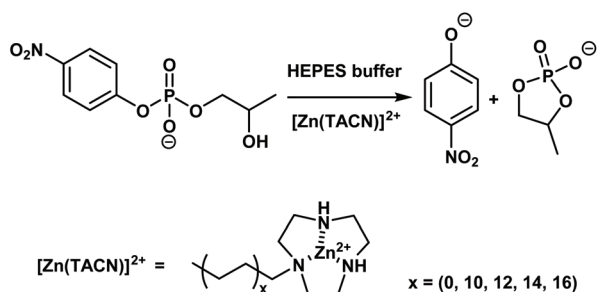
5.1.2 Micellar structures to enable organic chemistry in water. One of the major limitations to the use of water as a solvent for organic chemistry is the insolubility of many organic molecules and catalysts or their incompatibility with aqueous conditions. While the insolubility of some reagents does not always preclude the use of water as a reaction medium,¹⁴⁹ in some cases it leads to very low reactant or catalyst concentrations which hinders reaction rates. In addition, high dilution conditions are occasionally required to avoid the

formation of intractable slurries.¹⁵⁰ These requirements detract from the potential environmental benefits of using water as an alternative to volatile organic solvents. The use of micelles to solubilize many otherwise hydrophobic compounds allows for many of these limitations to be overcome and has been a key focus of recent work in micellar catalysis.^{117,118} When micellar catalysis conditions are used, there is often the need to use organic solvents to extract the reaction product(s) from the aqueous micellar solutions. Emulsification of the resultant extraction system can also be problematic, particularly if high surfactant concentrations are used.¹⁵¹ Selected examples of cases where micellar catalysis has been used in place of organic solvents will be highlighted below along with a discussion of the separation processes used and the impact of this on the generation of waste.

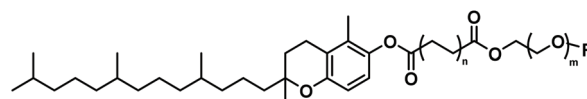
The effect of micellar catalysis on the total process waste has been explored in the field of transition metal catalyzed cross-coupling reactions.¹⁵² These reactions all featured the use of designer surfactants prepared from a tocopherol (vitamin E) moiety (Scheme 21). Heck, Sonagashira, and Suzuki-Miyauri couplings were compared with published industrial processes (Scheme 22) and it was found in all cases that the *E*-factors were significantly reduced by the use of the designer surfactant in water, even accounting for aqueous waste. The low *E*-factors were due to the ease of extraction of reaction products using small volumes of added organic solvent, the minimization of any aqueous workup steps, and the inherently higher reaction yields obtained under micellar catalysis conditions. The overall process is also rendered less hazardous from the use of the micellar catalysis conditions as the waste that is generated largely involves contaminated water and ethyl acetate, which is a relatively benign organic solvent.

Similarly, low *E*-factors have been found under closely related conditions for S_NAr reactions^{153,154} and the trifluoromethylation of heterocycles.¹⁵⁵ In addition to reduced waste, these processes could be performed under aqueous conditions avoiding the need for hazardous dipolar aprotic solvents for the S_NAr reactions and chlorinated solvents for the trifluoromethylation.

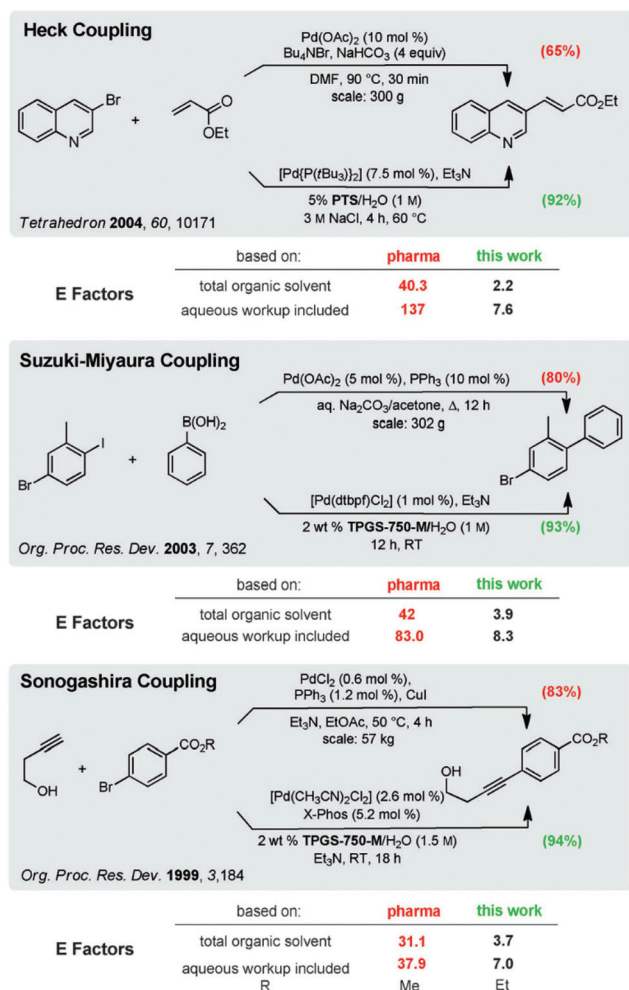
The design of self-assembled systems with product separation as a focus has also been investigated by Aubry and co-workers.¹⁵⁶ They studied the use of a quaternary ammonium molybdate surfactant for the generation of microemulsions for the peroxidation of alkenes (Scheme 23). These surfactants form a 3-phase system consisting of oil and water layers as well as a microemulsion phase. The aqueous phase contained the H₂O₂ oxidant while the oil phase initially contained reactants and allowed for the easy separation of products. The micro-



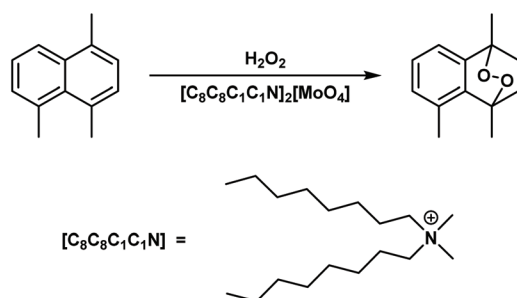
Scheme 20 Transphosphorylation of 2-hydroxypropyl-4-nitrophenyl phosphate catalysed by zinc TACN complexes, as investigated by Solís Muñana *et al.*¹⁴⁸



Scheme 21 Designer surfactants prepared by Lipshutz and coworkers. PTS has $n = 4$, $m = 14$, $R = H$; TPGS-750-M has $n = 1$, $m = 17$, $R = Me$.¹⁵²



Scheme 22 Comparison of *E*-factors for transition metal cross-coupling reactions performed in organic solvents and those conducted under micellar catalysis conditions.^{153–155} Reprinted with permission from ref. 152. Copyright 2013 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.



Scheme 23 Peroxidation of alkenes catalyzed by a microemulsion formed using a tetralkylammonium molybdate surfactant as described by Aubry and coworkers.¹⁵⁷

emulsion phase was the active phase. In this phase $[MoO_4]^{2-}$ catalytically generates singlet oxygen (1O_2) from the H_2O_2 leading to its reaction with the organic substrate. The use of a pure microemulsion phase using sodium dodecylsulfate com-

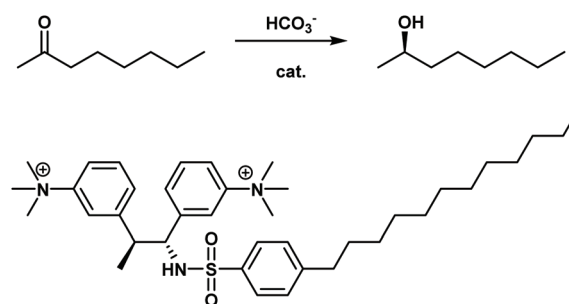
bined with sodium molybdate leads to similar catalytic activity however requires larger concentrations of amphiphile due to the need for a co-surfactant, leading to difficulties with the subsequent extraction of reaction products.¹⁵⁷

These examples illustrate that through appropriate design and selection of the surfactant and reaction conditions employed, micellar catalysis can be used to facilitate the transition from organic solvents to water as a reaction medium. For micellar catalysis to be genuinely useful as an approach in green chemistry, such considerations are essential.

5.1.3 Micellar catalysis to minimize transition metal use.

Minimizing the use of rare transition metal catalysts has attracted increasing attention over recent years due to concerns over resource depletion.¹⁵⁸ The incorporation of these metals into reaction products also represents a significant challenge for products to be used as active pharmaceutical ingredients.¹⁵⁹ As previously discussed, the effective local concentration of reactants and catalysts can be substantially increased within the micellar pseudophase relative to the bulk so long as these compounds are sufficiently hydrophobic or contain the appropriate charge to be adsorbed within this pseudophase. This provides the opportunity to use lower concentrations of transition metals as catalysts if the metal and substrate are strongly co-located.

One approach has been the incorporation of the active transition metal catalyst as a component of the surfactant through the preparation of metallomicelles or the use of inorganic counterions.¹⁶⁰ Li *et al.* explored the use of chiral Rh TsDPEN catalysts containing amphiphilic ligands for the asymmetric reduction of aliphatic ketones (Scheme 24).¹⁶¹ These reductions were facilitated by hydrogen transfer from HCO_3^- anions, so the micelles played the role of increasing the local concentration of both these anions and the Rh catalyst. Enantiomeric excesses (ee) of 90% could be achieved using these amphiphilic catalysts relative to only 37% for the same catalyst with a non-amphiphilic ligand at the same catalyst loading (1 mol%). The conversion and ee were observed to decrease sharply at temperatures above 40 °C which was inferred to be due to the breakdown of micellar structures, highlighting the need for the formation of aggregates to enable catalytic activity. The improved performance of the amphiphilic ligand demonstrates that reduced metal loadings



Scheme 24 Asymmetric ketone reduction reaction and amphiphilic TsDPEN ligand used by Li *et al.*¹⁶¹

can be used to achieve similar catalytic performance when the catalyst and reactants are confined within a self-assembled aggregate.

Lipshutz and co-workers have developed an approach for the use of very low transition metal loadings for an array of transition metal-catalyzed processes.¹¹⁸ This relies on the use of hydrophobic but not amphiphilic ligands in conjunction with a designer surfactant, which facilitates the solvation of the catalyst within the micellar interior. Three generations of ligands have been developed and successfully employed for the conduct of Pd catalyzed Suzuki–Miyaura couplings under micellar catalysis conditions (Scheme 25).^{162–164} These ligands enabled the use of low (<0.5 mol%) catalyst loadings, rather than typical loadings of 2–5 mol%. They also resulted in decreased Pd contamination in the resultant product to levels below FDA requirements, which would reduce solvent use for downstream purification processes. The three generations of ligands were designed to minimize the environmental impact of the catalyst. EvanPhos, the 2nd generation, was prepared to overcome the lengthy synthetic procedure required for HandaPhos, but it led to slightly compromised activity. N2Phos, the 3rd generation, was found to provide comparable activity to HandaPhos without the synthetic complexity and increased the scope of the Suzuki–Miyaura couplings to include the use of aryl chlorides as well as bromides and iodides as coupling partners. It was also illustrated that use of Fe nanoparticles derived from FeCl₃ containing <500 ppm Pd was able to achieve similar cross-coupling performance under micellar catalysis conditions.¹⁶⁵ This general approach could be conducted under continuous flow conditions and was extended to Sonagashira couplings.^{166,167} This highlights the versatility and likely industrial viability of this approach alongside the potential to markedly reduce the use of rare transition metal catalysts through micellar catalysis.

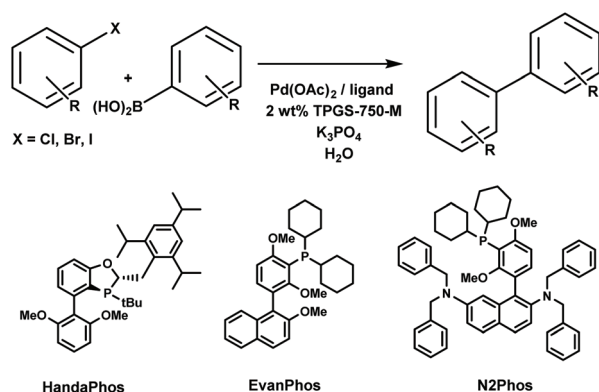
5.1.4 Micellar catalysis for incompatible reactions.

Nanoconfinement within micellar structures provides an alternative local reaction environment to bulk water. Remarkably, confinement within these micellar structures has facilitated the conduct of water-sensitive reactions despite the

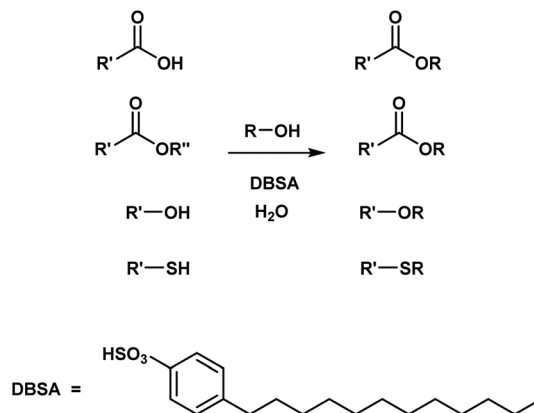
presence of the aqueous phase. Avoiding the need for rigorously dry solvents and reagents reduces the production of waste in the form of expended drying agents and energy for the distillation of solvents. The use of these nanoreactor-type environments has also allowed for the utilization of traditionally incompatible reagents if their confinement within separate regions of the solution can be achieved. One of the earliest examples of the conduct of a water-sensitive reaction under micellar catalysis conditions was the use of dodecylbenzene-sulfonic acid (DBSA) as a surfactant for a series of dehydration reactions in water (Scheme 26).¹⁶⁸ The surfactant acted as an acid catalyst and also led to the formation of micelles which sequestered the hydrophobic reaction products away from the aqueous phase, preventing their hydrolysis. This approach was successfully applied to esterification, etherification, thioetherification, and thioesterification. The use of non-amphiphilic strong acids such as sulfuric acid or *p*-toluenesulfonic acid resulted in vastly reduced yields. The equilibrium product yields obtained under neat conditions in the absence of water and under micellar catalysis conditions were similar, illustrating that water is effectively excluded from the micellar interior.

Krasovskiy *et al.* demonstrated the *in situ* generation of moisture-sensitive organozinc reagents within polyoxyethanyl *a*-tocopheryl sebacate (PTS) micelles from zinc powder and alkyl halides to perform Negishi couplings with aryl bromides.¹⁶⁹ The presence of the hydrophobic micellar interior was required for the stability of the organozinc reagent. The reaction in water without the surfactant or the addition of the preformed organozinc reagent producing significantly reduced yields. Of note is that the reaction involving these moisture-sensitive intermediates was more rapid under micellar catalysis conditions in water than in dry THF due to the increased local concentration of reactants and catalysts.

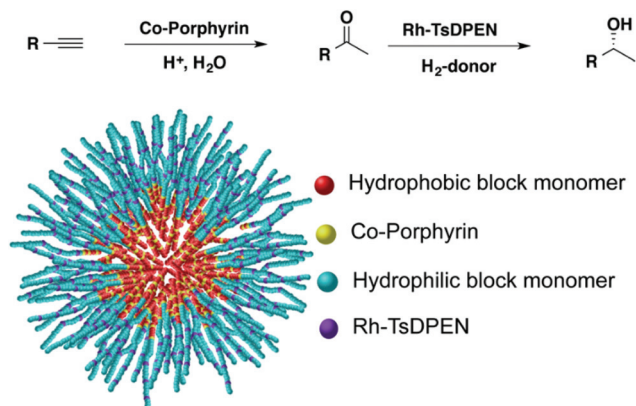
Lu *et al.* designed a surfactant based on the core-shell concept which contained a Co-porphyrin core with a Ru-TsDPEN shell to catalyze otherwise incompatible alkyne hydration and asymmetric transfer hydrogenation reactions (Scheme 27).¹⁷⁰ The surfactant was based on an ABC triblock



Scheme 25 General Suzuki–Miyauri coupling reaction and ligands used by Lipshutz and coworkers.¹¹⁸



Scheme 26 Dehydration reactions performed by Manabe *et al.* in water using DBSA.¹⁶⁸



Scheme 27 Core-shell surfactant and model reaction used by Lu *et al.*¹⁷⁰ Reprinted with permission. Copyright 2015 American Chemical Society.

copolymer that was functionalized with thiols, enabling the Co-porphyrin to be attached through a thiol-ene click reaction. The copolymer also contained a carboxylic acid side-chain to enable peptide coupling of the amine substituted Rh-TsDPEN catalyst. In the absence of surfactant, the Co-porphyrin and Rh-TsDPEN catalysts were inactive when combined while they could afford the relevant individual transformations when performed separately. In the presence of the core-shell surfactant, 95% conversion of phenylacetylene could be achieved with an ee of 96% for the alcohol product which compared favourably to the 60% conversion obtained under the stepwise addition of catalysts.

These examples demonstrate that water sensitivity does not necessarily preclude a reaction from being performed under micellar catalysis conditions if the confinement of the reactant, intermediate, or catalyst within the hydrophobic micellar interior can be achieved. There is also the possibility to use confinement within the micellar and aqueous pseudophases or within more complex self-assembled structures as a means of preventing incompatible catalysts or reagents from interacting, thereby facilitating multistep processes without the need for stepwise addition or additional separation and purification procedures.

5.1.5 Stimuli-responsive micellar catalysis. An exciting possibility within the field of micellar catalysis is the potential to develop stimuli-responsive catalysts. If it is possible to selectively 'switch' a catalyst from a dormant to an active state, then this will allow for the control of reactivity within the solution. This approach has the potential to facilitate complex multistep syntheses in a single pot by the appropriate switching of catalysts into their active or dormant states as the reaction proceeds, removing the need for the separation and purification of reactants and products after each catalytic step.

At the core of the concept of stimuli-responsive catalysis is the idea of switchable or stimuli-responsive surfactants.^{171,172} These are surfactants that change their critical aggregation concentration in response to an external stimulus such as

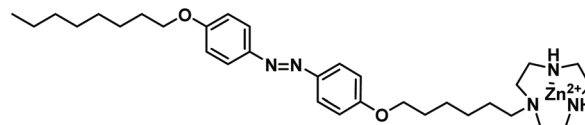


Fig. 19 Photoswitchable azobenzene derivative of Zn(TACN) used for stimuli-responsive micellar catalysis by Ren *et al.*¹⁷⁸

light,¹⁷³ temperature,¹⁷⁴ pH,^{171,175} magnetic fields,¹⁷⁶ or electrical potentials.¹⁷⁷ While many examples of these types of stimuli-responsive surfactants currently exist, surprisingly few have been explored for micellar catalysis.

In a recent example, Ren *et al.* explored the use of an azobenzene linker attached to a Zn(TACN) complex (Fig. 19) for the catalytic transphosphorylation of 2-hydroxyphenyl-4-nitrophenyl phosphate.¹⁷⁸ The azobenzene linker underwent an *E* to *Z* isomerization on exposure to 365 nm light which could be reversed on exposure to white light. The critical aggregation concentration of the *E* isomer was 20 μM while for the *Z* isomer it was 60 μM . It was demonstrated that at a catalyst concentration of 20 μM the use of 365 nm light enabled the down-regulation of catalytic activity which could be reversed on exposure to white light. While the magnitude of the decrease in rate was only a factor of 2, this does highlight the potential to use an external stimulus to control catalytic activity under micellar catalysis conditions.

Photoresponsive micellar catalysts have also been explored by Tang *et al.* for the oxidation of sulfides in water using a Ti-salen metallomicelle.¹⁷⁹ These surfactants formed hydrophobic nitrospiran groups on exposure to visible light, which led to the precipitation of the surfactant from the solution. In the dark, the hydrophilic merocyanine chain reformed, leading to the regeneration of the micelles. While the effect of visible light on catalytic activity was not directly investigated, the presence of micelles increased the turnover frequency relative to the Ti-salen complex in the absence of surfactant by a factor of 2, which implies that the activity on exposure to visible light would be reduced.

These examples highlight that despite the dearth of literature regarding the use of these stimuli-responsive surfactants for micellar catalysis, there remains enormous scope to harness these systems for more sophisticated catalytic processes. Micellar catalysis has successfully been used to create novel reaction environments. The combination of these advances with stimuli-responsive systems opens up the ability to move beyond enzyme-mimetics towards systems that more closely model cellular processes with interrelated multistep processes controlled by the appropriate signaling. Clearly, there is enormous potential scope for this type of advanced reaction control in water to play a significant role in the green chemical processes of the future.

5.2. On-water catalysis

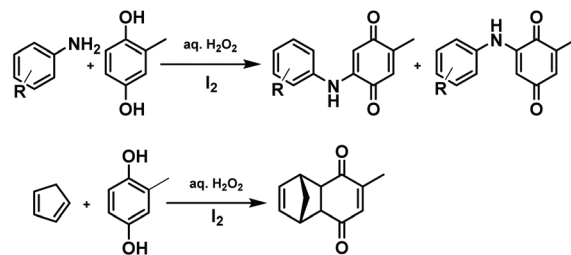
The dispersion of hydrophobic fluids in water where the reactants are confined within the dispersed phase has led to some

unusual outcomes due to the nature of the oil/water interface. Unlike systems operating under micellar catalysis conditions, no surfactant is required. However, continuous agitation is needed due to the thermodynamic instability of the dispersion.

Breslow and co-workers discovered that the use of water as a solvent for the Diels–Alder reaction led to increased reaction rates and selectivity for the *endo* products.^{180,181} Increasing the solubility of reactants in water through the addition of guanidinium chloride led to reduced reaction rates and selectivity, which suggested that the reaction was driven by hydrophobic effects. Narayan *et al.* identified that the use of aqueous dispersions can lead to increased reaction rates for the Diels–Alder reaction between quadricyclane and dimethyl azodicarboxylate.¹⁸² Moreover, the addition of methanol to the water phase did not affect the reaction rate until the system became homogeneous, when an approximately 10-fold decrease in rate could be observed. This demonstrated that the confinement of reactants within an immiscible hydrophobic phase was required for the rate enhancement to be observed.

The mechanism of this rate enhancement was the subject of a series of investigations over the subsequent decade.^{119,120,183,184} There now appears to be general agreement that there are two different mechanisms that can lead to rate enhancements of molecules within the oil phase near the oil/water interface. The first is due to an increase in hydrogen bond strength at the interface due to the presence of “dangling” hydrogen bonds created by the lack of a water molecule acting as a hydrogen bond acceptor.¹²⁰ The second mechanism involves a formal proton transfer from water bound to the interface due to the preference for OH[−] adsorption at the water/oil interface, which renders neutral water at this interface more acidic.¹¹⁹ The dominant mechanism is dependent on the basicity of the substrate, with more basic substrates typically undergoing acid-catalyzed reactions with the hydrogen bond effect more pronounced for less basic substrates.¹⁸⁴ As the interface is acting as a catalyst in both cases, the reaction rate will increase with the surface activity of the substrate and the relative surface area of the interface. This effect is also insensitive to the nature of the oil phase with on-water catalysis even able to occur within dispersed hydrophobic IL phases.¹⁸⁵

The ability to use the oil/water interface as a catalyst for substrates confined within the oil phase has subsequently been explored for a range of synthetic applications.¹⁸⁶ Norcott *et al.* described the use of on-water catalysis to prepare an on-water, in-water domino cascade.¹⁸⁷ Using aqueous H₂O₂ they explored the oxidation of toluhydroquinone and its coupling with substituted anilines (Scheme 28). Toluhydroquinone is oxidized to form the water-immiscible toluquinone which then undergoes the on-water catalyzed conjugate addition to the aniline. The Diels–Alder reaction of cyclopentadiene with the toluquinone under on-water conditions was also demonstrated although the desired domino reaction combining both in-water oxidations and the on-water conjugate addition and cycloaddition steps did not occur due to the formation of the



Scheme 28 In-water, on-water domino reactions studied by Norcott *et al.*¹⁸⁷

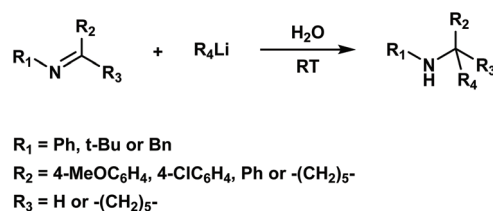
single addition products and an epoxide formed from the oxidation of the Diels–Alder addition product by H₂O₂. Nonetheless, this example illustrates the potential to use on-water catalysis and the partitioning of solutes between the aqueous and hydrophobic regions as a means to achieve one-pot multistep chemical syntheses.

An interesting recent example of the use of on-water catalysis involves the addition of organolithium reagents to hydrophobic unactivated imines (Scheme 29).¹⁸⁸ The yield of the desired amine was improved by the dispersion of the hydrophobic imine using active stirring prior to the addition of the organometallic reagent. Gentle stirring which led to reduced interfacial surface areas resulted in notably reduced yields, as did the use of D₂O which suggested the presence of an acid-catalysed on-water mechanism. This on-water approach enabled these additions to be performed in the absence of volatile organic solvents at room temperature in air as opposed to the typical inert, low temperature conditions employed within volatile organic solvents.

These examples demonstrate that increasing the interfacial surface area between hydrophobic and aqueous phases, where reactants are solvated in the hydrophobic phase, offers the potential for hydrogen-bond and acid-induced catalysis of a range of diverse reactions. In these cases, the reactants can be considered to be confined within the hydrophobic phase due to their low solubility in water with the “walls” of this confined phase acting as the catalyst.

5.3. Confinement in structured fluids

Another approach to using self-assembly for the nanoconfinement of reactants has been the use of inherently structured fluids as solvents. These fluids include ILs containing amphi-



Scheme 29 Addition of organolithium reagents to unactivated imines on-water, as studied by Dilauro *et al.*¹⁸⁸

philic nanostructures as well as more highly ordered liquid crystalline phases. The effect of nanoconfinement on reactions within these liquids can be broadly broken down into two different classes. One is the concentration or separation of reacting species on the basis of polarity, with parallels to micellar catalysis. The other involves the inherent order of the medium imparting unusual reaction outcomes. Both approaches offer the potential to achieve improvements in reaction rates and selectivities if these effects can be fully understood and utilized. Another emerging class of structured fluids is porous liquids, and the ability of this novel class of liquids to facilitate confinement-based catalysis will also be discussed.

5.3.1 Reactions in liquid crystals. Liquid crystals are anisotropic fluids.¹⁸⁹ Liquid crystals can be classified as thermotropic or lyotropic depending on whether the anisotropy and/or fluidity is induced due to temperature or added solvent respectively.¹⁹⁰ Lyotropic liquid crystalline phases are association colloids and reactivity within these fluids is classed as micellar catalysis as has previously been discussed. Thermotropic liquid crystals can be further divided into four distinct classes depending on the degree of organization. These classes are nematic, cholesteric, smectic, and columnar. Nematic and cholesteric phases both feature a preferred molecular orientation but the random molecular position with cholesteric phases being twisted through the medium with a certain periodicity rather than aligning in a single direction.¹⁹⁰ Smectic phases contain molecules co-located within a plane with several subclasses (smectic A, B, and C) based on the relative alignment of molecules within that plane. Columnar phases typically contain disc-shaped molecules which stack into columns and are classified on the relative arrangement of the columns with the most common arrangements being hexagonal, rectangular, and herringbone.

Unlike conventional solvents, thermotropic liquid crystals influence chemical reactions due to the effects of mesophase structure as well as solvent–solute interactions.¹⁸⁹ In other words, by confining reactants within the thermotropic mesophase, the relative energy of reactants and transition states is influenced by their size and shape relative to the solvent molecules as well as by electrostatic interactions. These effects are similar to some of the geometric constraints discussed previously with respect to confinement within pores. However, for most thermotropic liquid crystals derived from non-polar organic molecules, the weak intermolecular interactions lead to local disruption of these mesophases during the reaction with only subtle effects on reactivity, apart from cases where the reactants closely resemble the solvent molecules.¹⁹¹

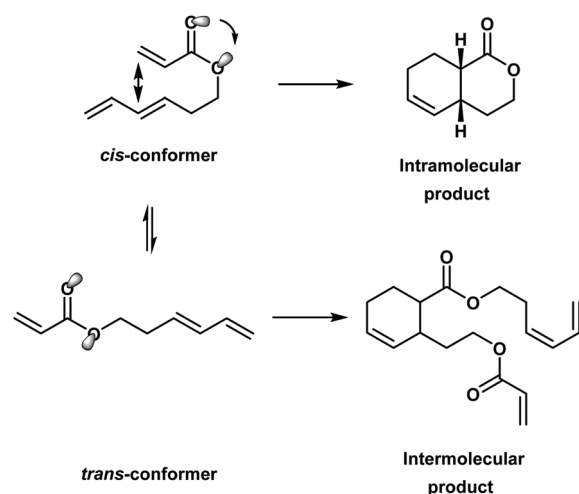
Recently, the use of ionic liquid crystals (ILCs) derived from IL-like compounds has been explored towards overcoming some of the limitations of traditional neutral thermotropic liquid crystals. ILCs have stronger solvent–solvent interactions and mesophases that can be stable over large temperature ranges which reduces the likelihood of solutes affecting the solvent ordering.¹²¹

Do and Schmitzer developed novel ILCs consisting of amphiphilic imidazolium groups on the periphery of a

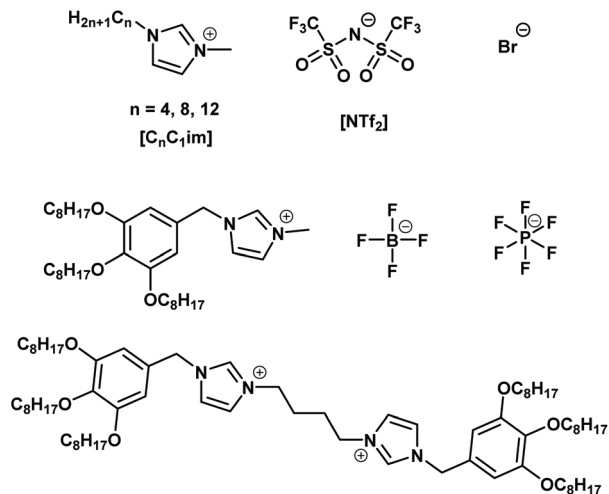
naphthalene or benzylic core. These IL crystals favoured intramolecular over intermolecular Diels–Alder reactions of a 1,3,9-decatriene derivative, even at high substrate concentrations (Scheme 30).^{192,193} This was compared to an IL, $[C_4C_1im][NTf_2]$, which showed a significant decrease in the isolated intramolecular product at high reactant concentrations. The favouring of the intramolecular reaction within the ILC was attributed to the organization of the bulky solvent around the solute, reducing the frequency of bimolecular collisions. Given this system was only compared with a single IL, it is difficult to ascertain the influence of other factors such as viscosity which may also influence the relative rates of unimolecular and bimolecular processes.

Bruce *et al.* explored the outcome of the Diels–Alder reaction between cyclopentadiene and methyl acrylate within 2 different classes of ILCs based on imidazolium cations and within an array of imidazolium ILs with varying degrees of amphiphilicity (Scheme 31), including $[C_{12}C_1im][BF_4]$ which forms a liquid crystalline smectic A phase.¹²¹ The *endo:exo* ratios obtained varied subtly with the nature of the solvent, with a general decrease with increasing amphiphilicity of the solvent. The *endo:exo* ratio was found to increase between $[C_4C_1im][NTf_2]$ and $[C_4C_1im][BF_4]$ but decrease between $[C_{12}C_1im][NTf_2]$ and $[C_{12}C_1im][BF_4]$ which was attributed to the reduced availability of imidazolium cation for hydrogen-bonding interactions within the liquid crystalline phase. It should be noted, however, that ILs containing $[C_4C_1im]^+$ cations typically do not display pronounced amphiphilic nanostructures whereas ILs with longer alkyl chains do. Hence this difference may also occur due to the greater solvophobic exclusion of hydrophobic reactants from the polar regions within the $[C_{12}C_1im][BF_4]$ IL, an effect which has been recently reported for related ILs.¹⁹⁴

Therefore, while confinement of reactants within liquid crystalline phases offers the potential to use the inherent order



Scheme 30 Competition between intra- and intermolecular Diels–Alder products studied by Do and Schmitzer. Reproduced from ref. 192 with permission from the Royal Society of Chemistry.



Scheme 31 Structures of the ILC and ILs studied for the Diels–Alder reaction between cyclopentadiene and methyl acrylate by Bruce *et al.* along with abbreviations used in the text.¹²¹

and anisotropy of the solvent to provide control over reaction outcomes, very few unambiguous examples of these effects have been demonstrated. For ILCs there is also the possibility that some of the effects described can be more readily attributed to the inherent IL nanostructure rather than specific effects arising from the anisotropy of the ILC phase. These nanostructure effects will be discussed in more detail below.

5.3.2 Reactions in nanostructured ionic liquids. ILs that contain at least one amphiphilic ion can form amphiphilic nanostructures with well-defined polar and non-polar domains.^{195–198} Solutes have been found to partition between these domains depending on their polarity and the presence of specific interactions between the solute and the IL ions.^{194,199,200} Despite increasing awareness of the existence of these structural features in ILs, surprisingly few examples of the effect of these nanostructures on chemical reactivity can be found in the literature and so the influence of these nanostructures remains relatively poorly understood. It is worth distinguishing between reactions conducted within ILs diluted only by small amounts of the reactants and those performed in the presence of considerable quantities of cosolvents such as water. While there have been some interesting developments in the latter,^{201–203} these systems are best described in terms of conventional micellar catalysis as described in section 5.1. The discussion below will focus on selected examples where the inherent nanostructure of the IL has been implicated in the reaction outcome, in addition to those discussed in section 5.3.1.

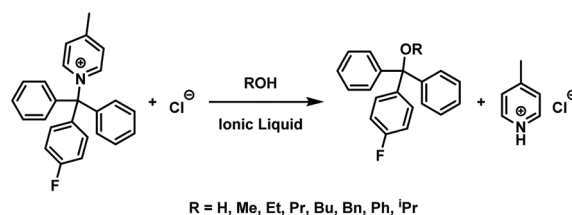
Puttick *et al.* explored the free radical polymerization of methyl methacrylate within a range of imidazolium ILs containing different anions and alkyl chain lengths.²⁰⁴ These studies were motivated by the observation that free-radical polymerization in ILs tended to result in higher molecular weight polymers and greater control than could typically be achieved within organic solvents. Using ROESY NMR experiments they identified the partitioning of the growing methyl

methacrylate polymer chain into the non-polar regions of the liquid. The separation of these growing chains from the radical initiator and smaller oligomers was attributed to the reduced rates of termination and increased molecular weights observed.

Weber *et al.* examined a nucleophilic substitution reaction featuring a cationic electrophile with an array of hydroxylic nucleophiles (Scheme 32).^{205,206} They discovered that the rate of reaction increased with the length of the alkyl chain of the IL with the greatest rate enhancements observed for the most polar nucleophiles. This effect was able to be quantitatively related to the volume of the nanostructured domains using a pseudophase model derived from the micellar catalysis literature, which highlights the parallels between these nanostructured ILs and micellar catalysis.

While these investigations have highlighted that the nanostructure of ILs can influence chemical reactivity, the scope of these effects remain relatively undefined. It appears that the greatest effects are seen for reactions that do not feature any significant charge development on progression from the reactants to the transition state. This is due to the known reorganization of ILs around charged solutes,²⁰⁷ which would lead to a breakdown in the local nanostructure surrounding the solute. While this potentially limits the use of nanostructured ILs to a select group of reactions, it does present the opportunity to use the inherent nanostructure of these liquids and the partitioning of solutes within them to facilitate control over chemical reactions. It also offers the possibility of performing otherwise incompatible reactions by confining reactants or catalysts within different regions of the IL although these avenues have not yet been explored.

5.3.3 Confinement in porous liquids. Porosity is a feature often associated with solids, which has found applications in different fields such as the confinement of catalytic systems, as discussed in section 3. The presence of permanent microporosity in liquids, however, was only suggested by the group of Stuart James in 2007.²⁰⁸ The existence of porous liquids opens avenues for combining the advantages of porous systems with the fluidity of liquids. In the context of catalysis, one can envisage the confinement of species in nanoreactors that are well-defined, sturdy, and permanent, like in solid pores and encapsulated systems; while also overcoming mass transfer limitations associated with the solid state, like in self-assembled structures.



Scheme 32 Nucleophilic substitution reaction used by Weber *et al.* to investigate the effect of nanostructured ILs on reactivity.^{205,206}

Porous liquids were conceptualized as consisting of molecules in the liquid state that contain empty voids or by porous particles or networks that are dispersed within a liquid phase. The pores must be permanent, well-defined, and large enough to host molecules. Three types of porous liquids have been proposed (Fig. 20): type 1 consists of neat liquids, formed by molecules that contain an internal cavity and whose structures are not flexible enough to fold and occupy their own pores or free spaces in the vicinity; type 2 is formed by empty hosts dissolved in solvents that are bulky enough to be excluded from the interior of the cavities; and type 3 consists of solid microporous frameworks suspended in sterically hindered solvents.

Despite the conceptualization in 2007, real-life examples of porous liquids were not available until the following decade.²⁰⁹ Types 1,^{210–212} 2^{213,214} and 3²¹⁵ porous liquids have now been reported. To date, a major application of these liquids has been the adsorption of small gas molecules. The major energetic cost for the dissolution of gases in liquids is the formation of cavities for the solute, hence the important role of the pre-formed cavities in gas adsorption. Interestingly, Greenaway *et al.* were able to unload dissolved CO₂ from imine-based type 1 porous liquids by sonication. This could be repeated five times without loss of adsorption capacity,²¹⁴ uncovering the potential for the development of, for instance, recyclable catalytic systems.

Nonetheless, only recently the Maschmeyer group brought this concept to the field of catalysis. They used a strategy developed by Zhang *et al.* to produce a type 1 porous liquid from porous hollow silica nanospheres.²¹² The particles were functionalized with a corona of organosilanes and the anions exchanged for poly(ethylene glycol)-tailed sulfonate (PEGs) tails (called canopy structures), which turned the solid particles into a fluid porous material (Fig. 21). The Maschmeyer group encapsulated Au, Pt, and Pd nanoparticles in a rattle-type fashion inside the silica nanoshells and adapted Zhang's methodology to turn them into a porous liquid.²¹⁶ In a subsequent study, the Maschmeyer group used the Pt-containing porous liquid for the hydrogenation of 1-decene under mild conditions (40 °C, 1 atm of H₂).²¹⁷ They compared the per-

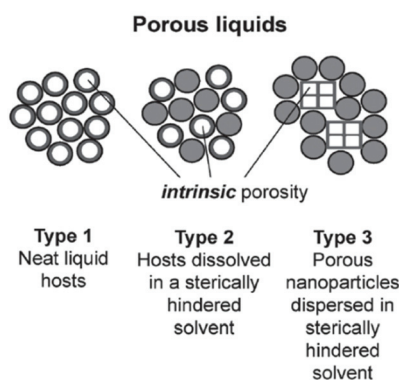


Fig. 20 Types of porous liquids. Reprinted with permission from ref. 209. Copyright 2016 WILEY-VCH Verlag GmbH & Co. KGaA, Weinheim.

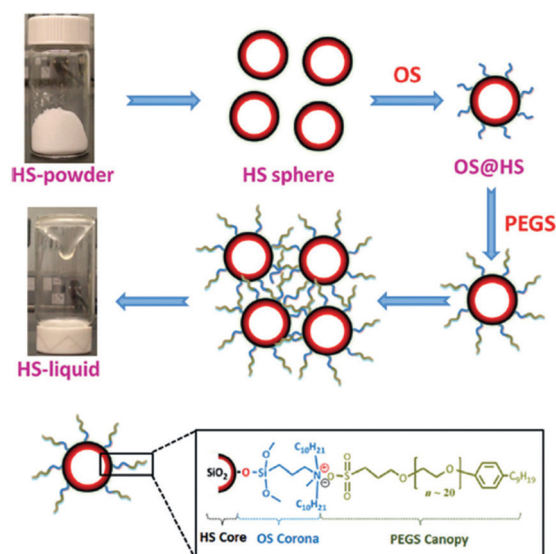


Fig. 21 Synthetic strategy for porous liquid fabrication. HS = hollow silica, OS = organosilane. Reprinted with permission from ref. 212. Copyright 2015 Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim.

formance of different catalytic systems to examine the importance of the porous liquid and its key structural features. Although all systems were effective catalysts, unfunctionalized silica nanoshells were the most effective, while the neat porous liquid was the least efficient in terms of conversion over time. The comparison was however not perfect because Pt nanoparticles aggregated during the silica functionalization process, so the porous liquid had larger Pt nanoparticles than the control experiment. The neat porous liquid was also a more viscous reaction medium, which could have contributed to the slower reaction due to reduced diffusion rates. Nonetheless, the porous liquid could be reused 3 times without loss of activity and promoted the conversion of different substrates. An interesting point is that the neat porous liquid was able to convert 4-chloronitrobenzene below its melting point without the presence of a solvent, hence showcasing a potential advantage of such systems in the green chemistry context.

Porous liquids are novel structures that combine the fascinating ability of nanoconfinement found in porous materials with the fluidity and enhanced mass transfer of liquids. There is still much to discover in this field, but we foresee great potential to be uncovered with the use of such systems to catalyze greener transformations, particularly for reactions involving gaseous reactants.

6. Outlook/final remarks

The examples highlighted in the article showcase that confined systems can make more with less. Hence they are a valuable tool to create more sustainable reaction conditions.

It is apparent that the confinement of reactants, intermediates, and catalysts can lead to remarkable improvements in the efficiencies of an array of chemical transformations. Proof of principle examples have been highlighted here for systems where the confinement occurs within pores or cavities of solid porous materials, inside hollow solid materials, and within self-assembled systems. While the nature of confinement varies between these diverse systems, all demonstrate the importance of interfaces and the co-location of reactants, catalysts, and intermediates in governing improvements in reaction efficiency. The precise control over diffusion effects as well as the geometry of transition states are also powerful ways to achieve higher activities and selectivities.

Advances in the use of confinement for green chemistry will rely on a greater understanding of the roles of geometrical and electronic effects in different reaction systems. This understanding is challenging due to the uniqueness of each combination of reactants and catalysts, which makes generalization and predictions somewhat difficult to make. As demonstrated particularly on section 2 of this review, confinement effects are powerful but need to be designed for the peculiarities. Enzymes, for instance, are the perfect example of confined systems that can be achieved if reaction spaces and intermolecular interactions are optimized for a given catalytic process. Being able to design *a priori* such spaces will be a substantial step forward in the development of more reactive, selective, and specific catalytic systems. Thus, we see computational chemistry playing an invaluable role in the future of confined catalysis.

Systems confined within solid porous or hollow materials, such as the ones presented on sections 3 and 4, have demonstrated remarkable versatility in the context of confinement. Future work will likely focus on the synthesis of these materials themselves. Small scales and low reproducibilities are non unusual in the world of nanomaterial synthesis, specially for hierarchical structures. These represent challenges to the full understanding and optimization of processes. Similarly, successful large scale nanomaterial syntheses will be a key factor in the wide application of confined catalysis within porous and hollow structures.

On the other hand, self-assembled systems like the ones presented in section 5, are reproducible and scalable but recycling and recovery of materials is a key challenge. These will likely be addressed by the development of the ability to 'switch' the self-assembly to facilitate the recovery of products.

Overall, confinement provides an excellent tool for the high-level control of chemical reactions that cannot be provided within bulk homogeneous solutions. Understanding, harnessing, and utilising this effect will pave the way for a quantum leap in the development of more active and selective catalytic processes, a key cornerstone for green chemistry in the 21st century.

Conflicts of interest

There are no conflicts to declare.

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